

HCSM-00 (Tenth Revision)

Foundations, Notation, and Mapping

The Hodge Complex Standard Model: Substrate, Hodge Complex, Spectral Midpoint,
Processor, Output Structures, Emergent Physics, and Cosmology

Stanley Preschutti

Entropy Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

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Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, mixing matrices, spacetime dimension, graviton, Newton constant, and vacuum stability coefficient of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The substrate is selected by the unified substrate selection theorem. The emergent spacetime dimension is fixed by the topology of the Hodge-Dirac kernel. The Hodge complex of differential forms carries the operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation.

This paper establishes the framework's foundational objects and states the notation. All seven rank blockers of the framework's derivation program are closed. The framework has a single dimensionful anchor (the Planck mass M_P) and derives every other quantity from the substrate at $d = 4$ and $L = 8$.

The paper establishes the following theorems. The 5-point Laplacian is the unique D_4 -invariant nearest-neighbour positive-semidefinite operator with kernel equal to the constants. The Hodge-Dirac operator satisfies $D^2|_{\Omega^k} = \Delta \cdot \text{id}$ on each form degree, with smallest nonzero eigenvalue $2 \sin(\pi/L)$ on Λ_L . The Lorentzian metric on $\ker D$ is the unique D_4 -invariant symmetric form that vanishes on pure-degree forms and has signature $(3, 1)$. The Kernel-Diagonal Theorem yields the EFT coefficient $c = 5/9$, the mass-map coefficient $(d-1)/2 = 3/2$, and the kernel dimension $10 = (d-1)^2 + 1$ as three traces of one projector. The Λ -promotion mechanism is a pure projection onto the timelike direction of $\ker D$ and the refined dark energy density matches observation to 0.010%.

The charge sector is a rigidity-class result: five structural constraints reduce the charge assignment from $5^{14} \approx 6 \times 10^9$ configurations to exactly two, both with total charge $-1e$. The residual binary choice is resolved by the Midpoint Sorting Theorem of HCSM-23 Rev. 7.

The present Tenth Revision incorporates eight structural sharpenings. First, the processor commutant structure is integrated: the loop commutant $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ is a 3-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$, and the DNLS commutant $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ is 10-dimensional, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$ (HCSM-15). Second, the doubled-momentum pairing mechanism is integrated: the B_1 -compressed Hartree kernel vanishes identically, $K_{B_1} = 0$, by the explicit B_1 basis vectors constructed in Appendix A (HCSM-15). Third, the chirality-breaking localization is integrated: the processor commutator $[T, P_{rs}]$ is supported exactly on the two-element set $\{\text{dark}, \tau\}$ (HCSM-13). Fourth, the winding generator kernel structure is refined: the kernel- T decomposition is $(6, 1, 3)$, the family-block eigenvalues are $\pm L/2$, and the four parity-block indicators span a 4-dimensional subspace of the kernel (HCSM-14). Fifth, the class partition derivation is corrected: the class partition is the joint (D, T) spectrum (HCSM-05 Theorem 7.7), not a frequency decomposition of H_{wind} . Sixth, the M_{loop} notation is clarified: M_{loop} denotes the diagonal class operator (HCSM-17 Definition 2.1), while the five-layer composition is denoted $M_{5\text{layer}}$.

Seventh, the explicit B_1 basis vectors are constructed (Appendix A). Eighth, the mirror cancellation, $5 + 9$ split, Hodge-HCSM interface, and orientation-flip $\mathbb{Z}_2$ are integrated.

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1 Introduction

1.1 What HCSM is

The Hodge Complex Standard Model derives the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework's central claim is that the Standard Model is not a fundamental description of nature but the residue left when a small, discrete, information-bearing lattice cancels against its own mirror.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

HCSM does not ask what fields exist or what symmetry groups are broken. It asks a single question: what substrate could produce exactly the Standard Model, and nothing else? The answer is a discrete lattice whose mirror cancellation leaves exactly the observed particle content. The mirror-cancellation theorem of HCSM-10 [10] makes this statement precise: the Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, 50 of the 64 scalar modes pair into cancelling partners, and only the 14 self-paired modes at $\lambda = 4$ survive. The residual ratio is exactly $7/32$.

1.2 The single dimensionful anchor

The framework has exactly one dimensionful input: the Planck mass M_P . Every other dimensionful quantity is a derived multiple of an appropriate power of M_P . The dimensionless integers $d = 4$ and $N = 64$ are derived from the substrate (Section 2); the substrate size $L = 8$ is derived from the unified selection theorem (Section 2.6); and the emergent Lorentzian signature $(3, 1)$ is derived from the Hodge star on $\ker D$ (Section 3, Theorem 3.9).

At the particle level, the framework has zero empirical anchors. The identification of the 14 modes at $\lambda = 4$ with one Standard Model generation was until recently a disclosed input; it is now a theorem (HCSM-23 Rev. 7 [23]). The charge table was until recently a disclosed input up to a binary choice; it is now a theorem whose binary residue is resolved by the same sorting theorem.

As of the present revision, the gauge sector also has zero empirical anchors in the physical-identification step. The identification of the A_1 isotypic block with physical color $\mathfrak{su}(3)_c$ is a theorem (HCSM-06 [6], Theorem 5.4): the A_1 block is the unique three-dimensional D_4 -invariant subspace of the 14-mode multiplet, and the dimension-and-invariance argument forces this identification. The identification of the B_1 block with physical weak isospin is a labeling convention (HCSM-06 [6], Theorem 5.10), forced by the outer automorphism of D_4 .

1.3 Architectural levels

The framework's architecture is a chain from the substrate to observable physics, organized in seven levels.

1. **Level 0: The Substrate.** The periodic torus $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with the 5-point Laplacian. The substrate is selected by the unified substrate selection theorem from the single lattice primitive.
2. **Level 1: The Hodge Complex.** The exterior algebra $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ of differential forms with the Hodge-Dirac operator $D = d + d^*$, satisfying $D^2 = \Delta$ exactly on each form degree.
3. **Level 2: The Spectral Midpoint.** The self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian and its 14-mode eigenspace, carrying the D_4 isotypic decomposition $14 = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$.
4. **Level 3: The Processor and Its Dynamics.** The winding generator H_{wind} , the processor $T = I - P_C$, the class partition $\{A, B, C\}$, the loop operator M_{loop} , and the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$.

5. **Level 4: Output Structures.** The charge functional q_3 , the mass map $m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i)$, the $N/4$ anchors, and the Midpoint Sorting Theorem.
6. **Level 5: Emergent Physics.** The gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$, hypercharge quantization, flavor mixing (CKM and PMNS), the Higgs sector, and gravity.
7. **Level 6: Cosmology.** The kL crossover, dark energy, vacuum stability, and the Λ -promotion mechanism.

1.4 Continuum limit and emergent Lorentz invariance

The substrate is not a discretization of an underlying continuum. The framework's claim is the opposite: the continuum emerges from the substrate. The physical scale is set by the mass map and the kL crossover.

The 5-point Laplacian has the isotropic expansion

$$\lambda(n_1, n_2) = \frac{\pi^2}{16}(n_1^2 + n_2^2) + O(n^4). \quad (1)$$

The continuum limit of the Hodge complex is the subject of program items HCSM-68 through HCSM-71. It is the single open item outside the particle sector.

2 Level 0: The Substrate

2.1 The periodic $L \times L$ torus

The substrate is the periodic square lattice

$$\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L, \quad N = L^2, \quad (2)$$

consisting of N sites arranged in an $L \times L$ grid with periodic boundary conditions. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$; equivalently, the coordinates x_1 and x_2 are each defined modulo L . Topologically, Λ_L is the torus $T^2 = S^1 \times S^1$.

The D_4 point group of the square acts on Λ_L by the eight rotations and reflections of the lattice. Its presentation is $D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle$.

2.2 The 5-point Laplacian is forced

Theorem 2.1 (Laplacian uniqueness). *Let $\Delta : \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a linear operator on the substrate $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ satisfying:*

1. *D_4 -invariance: $[R, \Delta] = 0$ for every $R \in D_4$;*
2. *nearest-neighbour locality: $(\Delta f)(x)$ depends only on $f(x)$ and f at the four nearest neighbours of x ;*
3. *positivity and kernel: Δ is positive-semidefinite and $\ker \Delta$ is exactly the set of constant functions.*

Then Δ is uniquely determined up to an overall positive scale, and equals the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x + e). \quad (3)$$

Proof. Condition (2) writes $\Delta f(x) = a_0 f(x) + \sum_{\mu} a_{\mu} f(x + \mu)$. The four nearest-neighbour directions $\pm e_1, \pm e_2$ lie in a single D_4 orbit. Hence condition (1) forces $a_{\mu} = a$ for all four directions. Condition (3) forces $\Delta \mathbf{1} = 0$, so $a_0 + 4a = 0$, giving $a_0 = -4a$. Normalizing $a = 1$ gives the 5-point Laplacian. $\square$

Remark 2.2. The 5-point Laplacian annihilates the constants by construction. Its eigenvalue on the plane wave $\phi_{n_1, n_2}(x) = \exp(2\pi i(n_1 x_1 + n_2 x_2)/L)$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right), \quad (4)$$

with $n_1, n_2 \in \{0, 1, \dots, L-1\}$. The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$.

2.3 The framework relation $N = d \cdot 2^d$

The framework uses the relation

$$N = d \cdot 2^d, \quad (5)$$

with N the number of substrate sites and d the emergent spacetime dimension. At $d = 4$ this gives $N = 4 \cdot 2^4 = 64$. Since $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ requires $N = L^2$, this gives $L = \sqrt{64} = 8$.

At $d = 4$, $N = 64$ is a perfect square. The integer solutions of $d \cdot 2^d = L^2$ form the infinite family $d \in \{4k^2 : k \geq 1\}$, with $d = 16, 36, 64, \dots$ also giving perfect squares. The physical selection of $d = 4$ comes from the unified selection theorem of Section 2.6.

2.4 Two derivations of $L = 8$

The substrate size $L = 8$ is selected by two independent derivations.

First derivation: equal entropy spacing. On the $L \times L$ torus, the standing-wave modes fall into discrete Shannon entropy classes.

Theorem 2.3 (Equal entropy spacing selects $L = 8$). *Among all integers $4 \leq L \leq 10^4$, the only L for which the $L \times L$ torus has exactly three equally spaced Shannon entropy classes is $L = 8$. At $L = 8$ the three classes are 6.0 bits, 5.5 bits, and 5.0 bits, with uniform spacing 0.5 bits.*

Second derivation: maximal winding cancellation. The winding generator generalizes to any even L as $H_{\text{wind}}(L) = i \sin(\pi M/L)$ on the self-paired mode set of Λ_L .

Theorem 2.4 (Maximal cancellation selects $L = 8$). *Let $\mathcal{G}_L$ be the parity block graph of $H_{\text{wind}}(L)$ on the five nodes $\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$. The number of block couplings that vanish identically is*

$$\#\text{zeros}(\mathcal{G}_L) = \begin{cases} 3, & L = 4, \\ 4, & L \equiv 2 \pmod{4}, \\ 7, & L \equiv 0 \pmod{4}, \quad L \geq 12, \\ 10, & L = 8. \end{cases} \quad (6)$$

Among the two maximally-cancelling sizes, only $L = 8$ carries the 14-mode multiplet. At $L = 8$ the surviving five block couplings form a unicyclic graph on $d+1 = 5$ nodes with rank-1 edges; the winding kernel has dimension $d(d+1)/2 = 10$; and the kernel-diagonal values are $\{2/d, (d-1)/d\}$.

The two derivations are independent: the first uses only the torus Laplacian's entropy classes; the second uses only the winding generator's parity block structure.

2.5 The 14-mode count

Theorem 2.5 (Midpoint multiplicity). *On the $L \times L$ torus with L even, the multiplicity of the self-paired eigenvalue $\lambda = 4$ is*

$$\text{mult}_{\lambda=4}(L) = 2(L-1). \quad (7)$$

At $L = 8$ this gives 14.

Proof. The self-paired eigenvalue is the fixed point of the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$. Its eigenspace consists of the modes (n_1, n_2) satisfying $\cos(2\pi n_1/L) + \cos(2\pi n_2/L) = 0$. Using the sum-to-product identity $\cos a + \cos b = 2 \cos((a+b)/2) \cos((a-b)/2)$, this becomes $2 \cos(\pi(n_1 + n_2)/L) \cos(\pi(n_1 - n_2)/L) = 0$. The solutions are $n_1 + n_2 \equiv L/2 \pmod{L}$ or $n_1 - n_2 \equiv L/2 \pmod{L}$. Each congruence has L solutions; the two congruences intersect in exactly two points. By inclusion-exclusion, the total is $2L - 2 = 2(L - 1)$. $\square$

The 14 modes at $\lambda = 4$ on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ are the subject of HCSM-04 [4] and are identified with one Standard Model generation in HCSM-23 Rev. 7 [23].

2.6 Unified substrate selection

The substrate size $L = 8$ and the emergent dimension $d = 4$ are selected by four independent derivations in the current framework: equal entropy spacing on the torus Laplacian; maximal winding cancellation of the winding generator; Lorentzian signature on $\ker D$; and the Betti numbers of T^2 . This subsection shows that they are corollaries of a single selection theorem.

2.6.1 The two framework identities

Identity 1 (site-count relation). The number of substrate sites N and the emergent dimension d satisfy

$$N = d \cdot 2^d. \quad (8)$$

Identity 2 (mode-content relation). The self-paired mode count of the torus Laplacian at the midpoint eigenvalue equals the total particle content of one Standard Model generation:

$$\text{mult}_{\lambda=4}(L) = |F| + |B_{\text{bos}}|. \quad (9)$$

The right-hand side is derived content: the fermion-sector dimension $|F| = d(d+1)/2$ (HCSM-13 [13], HCSM-19 [19]) and the boson-sector dimension $|B_{\text{bos}}| = d$ (HCSM-13 [13], HCSM-19 [19]). At $d = 4$ they give $|F| = 10$, $|B_{\text{bos}}| = 4$, and total content 14.

The left-hand side is HCSM-04's multiplicity theorem:

$$\text{mult}_{\lambda=4}(L) = 2(L - 1). \quad (10)$$

2.6.2 The selection theorem

Theorem 2.6 (Unified substrate selection). *The system*

$$L^2 = d \cdot 2^d, \quad 2(L - 1) = d(d + 3)/2, \quad (11)$$

has the unique even-integer solution $L = 8$, $d = 4$, $N = 64$.

Proof. By Identity 1, $L = \sqrt{d \cdot 2^d}$. For L to be an even integer, $d \cdot 2^d$ must be a perfect square. Since 2^d is a perfect square iff d is even, write $d = 2m$. Then $d \cdot 2^d = 2m \cdot 2^{2m}$, which is a perfect square iff $2m$ is a perfect square, i.e. $m = 2k^2$ for integer $k \geq 1$. Hence

$$d = 4k^2, \quad L = 2^{2k^2+1} \cdot k. \quad (12)$$

The mode-content identity gives $2^{2k^2+1}k - 1 = k^2(4k^2 + 3)$.

Case $k = 1$: Left side $2^3 \cdot 1 - 1 = 7$. Right side $1 \cdot 7 = 7$. Satisfied.

Case $k = 2$: Left side $2^9 \cdot 2 - 1 = 1023$. Right side $4 \cdot 19 = 76$. Left exceeds right.

Case $k \geq 3$: The left side grows as $k \cdot 2^{2k^2+1}$, super-exponentially in k^2 ; the right side grows as $4k^4 + 3k^2$, polynomially. The left strictly dominates for all $k \geq 2$. Hence $k = 1$ is the unique solution, giving $d = 4$, $L = 8$, $N = 64$. $\square$

Corollary 2.7 (Uniqueness). *The pair $(L, d) = (8, 4)$ is the unique even-integer pair satisfying the two framework identities simultaneously.*

2.6.3 The four prior derivations as corollaries

Corollary 2.8 (Entropy spacing). *At $L = 8$, the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ has exactly three equally spaced Shannon entropy classes: 6.0, 5.5, and 5.0 bits.*

Corollary 2.9 (Maximal winding cancellation). *At $L = 8$, the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ has cancellation depth 10 on the five-node parity block graph, with kernel dimension $d(d+1)/2 = 10$.*

Corollary 2.10 (Lorentzian signature). *At $L = 8$, the Hodge star on $\ker D = H^0 + H^1 + H^2$ has eigenvalues $\{+1, -1, +i, -i\}$, giving a metric of signature $(3, 1)$ on $\ker D$.*

Corollary 2.11 (Betti numbers). *At $L = 8$, the kernel dimension is $d = \dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$.*

After the unified selection, the framework has exactly one open item outside the particle sector: the continuum limit.

2.6.4 What this changes

Item	Before	After
Derivations of $L = 8$	2 independent	1 theorem, 1 corollary
Derivations of $d = 4$	2 independent	1 theorem, 2 corollaries
Status of $L = 8$	4 consistent derivations	1 theorem, 3 checks
Status of $d = 4$	4 consistent derivations	1 theorem, 3 checks

2.7 The single lattice primitive

The mirror-cancellation theorem of HCSM-10 [10] sharpens the substrate-level axiom structure. Where HCSM-01 [1] lists three substrate-level disclosures, the mirror-cancellation theorem reduces these to a single primitive.

Axiom 1 (Lattice primitive). *There exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.*

Every object in the framework, including the substrate size $L = 8$, the 5-point Laplacian, the Hodge complex, the mirror symmetry, the 14-mode multiplet, and the residual $7/32$, is a theorem of Axiom 1. No physical input, no empirical anchor, and no fitted parameter is used at the substrate level.

2.8 Mirror cancellation

Definition 2.12 (Half-period translation). *The half-period translation is the map*

$$T_{\text{half}}(n_1, n_2) = ((n_1 + L/2) \bmod L, (n_2 + L/2) \bmod L). \quad (13)$$

Lemma 2.13. *T_{half} is an involution on $\{0, \dots, L-1\}^2$.*

Proof. $(n + L/2 + L/2) \bmod L = n \bmod L$ for both coordinates. □

Lemma 2.14 (Cosine shift). *For $n \in \{0, \dots, L-1\}$ with L even,*

$$\cos\left(\frac{2\pi \cdot ((n + L/2) \bmod L)}{L}\right) = -\cos\left(\frac{2\pi n}{L}\right). \quad (14)$$

Proof. Let $m = (n+L/2) \bmod L$. If $n < L/2$, then $m = n+L/2$ and $\cos(2\pi m/L) = \cos(2\pi n/L + \pi) = -\cos(2\pi n/L)$. If $n \geq L/2$, then $m = n - L/2$ and $\cos(2\pi m/L) = \cos(2\pi n/L - \pi) = -\cos(2\pi n/L)$. Both cases coincide. $\square$

Theorem 2.15 (Mirror symmetry). *For every momentum pair (n_1, n_2) ,*

$$\lambda(T_{\text{half}}(n_1, n_2)) = 8 - \lambda(n_1, n_2). \quad (15)$$

Consequently, the spectrum of Δ on Λ_L is exactly mirror-symmetric: if λ is an eigenvalue with multiplicity m , then $8 - \lambda$ is also an eigenvalue with multiplicity m .

Proof. By (4), $\lambda(n_1, n_2) = 4 - 2\cos(2\pi n_1/L) - 2\cos(2\pi n_2/L)$. Applying T_{half} to both coordinates and using Lemma 2.14:

$$\begin{aligned} \lambda(T_{\text{half}}(n_1, n_2)) &= 4 - 2(-\cos(2\pi n_1/L)) - 2(-\cos(2\pi n_2/L)) \\ &= 4 + 2\cos(2\pi n_1/L) + 2\cos(2\pi n_2/L) \\ &= 8 - \lambda(n_1, n_2). \end{aligned} \quad (16)$$

T_{half} is a bijection by Lemma 2.13, so it pairs eigenvalues λ and $8 - \lambda$ at equal multiplicity. $\square$

Corollary 2.16. *Every eigenvalue $\lambda \neq 4$ appears in a mirror pair $(\lambda, 8 - \lambda)$ of equal multiplicity.*

2.9 The $50 + 14$ decomposition and the residual $7/32$

Theorem 2.17 (Self-paired eigenvalue). *The unique self-paired eigenvalue is $\lambda = 4$.*

Proof. $\lambda = 8 - \lambda$ implies $2\lambda = 8$, hence $\lambda = 4$. $\square$

Corollary 2.18 ($50 + 14$ decomposition). *At $L = 8$, the 64 modes of Λ_8 decompose as*

$$50 \text{ pair modes} + 14 \text{ residual modes at } \lambda = 4. \quad (17)$$

Proof. Immediate from the preceding corollary and Theorem 2.5: $(64 - 14)/2 = 25$ pairs, hence 50 pair modes, plus 14 residual modes. $\square$

Theorem 2.19 (General cancellation). *Let $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $F(\lambda) + F(8 - \lambda) = c$ for all λ and some constant $c \in \mathbb{R}$. Then*

$$\sum_m F(\lambda_m) = c \cdot \frac{L^2 - 2(L - 1)}{2} + F(4) \cdot 2(L - 1). \quad (18)$$

Proof. Group the L^2 modes into $(L^2 - 2(L - 1))$ paired modes and $2(L - 1)$ residual modes at $\lambda = 4$. Each pair $(\lambda, 8 - \lambda)$ with multiplicity m_p contributes $m_p \cdot c$ by hypothesis. Summing over all pairs gives $c \cdot (L^2 - 2(L - 1))/2$. Adding the residual contribution $2(L - 1) \cdot F(4)$ gives the result. $\square$

Corollary 2.20 ($L = 8$). *At $L = 8$,*

$$\sum_m F(\lambda_m) = 25c + 14F(4). \quad (19)$$

Proof. Substitute $L = 8$: $(64 - 14)/2 = 25$; $2 \cdot 7 = 14$. $\square$

Theorem 2.21 (Residual ratio $7/32$). *For the zero-point convention $F(\lambda) = \lambda/2$ and $L = 8$:*

$$\sum_m F(\lambda_m) = 128, \quad \sum_{m \in \mathcal{R}} F(\lambda_m) = 28, \quad \frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}, \quad (20)$$

where $\mathcal{R}$ is the residual set of 14 modes at $\lambda = 4$.

Proof. Compute $\sum_m \lambda_m$ exactly. Using the trigonometric identity $\sum_{n=0}^7 \cos(\pi n/4) = 0$, we have

$$\begin{aligned} \sum_m \lambda_m &= \sum_{n_1, n_2=0}^7 \left[4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right) \right] \\ &= 4 \cdot 64 - 2 \cdot 8 \sum_{n=0}^7 \cos\left(\frac{\pi n}{4}\right) - 2 \cdot 8 \sum_{n=0}^7 \cos\left(\frac{\pi n}{4}\right) = 256. \end{aligned} \quad (21)$$

Therefore $\sum_m \lambda_m/2 = 128$. The residual contribution is $\sum_{m \in \mathcal{R}} \lambda_m/2 = 14 \cdot (4/2) = 28$. Ratio: $28/128 = 7/32$. $\square$

Theorem 2.22 (Convention independence). *Let $F(\lambda) = \alpha\lambda + \beta$ with $\alpha > 0$ and $F(4) \neq 0$. Then*

$$\frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}. \quad (22)$$

If $F(4) = 0$ (equivalently $\beta = -4\alpha$), both numerator and denominator vanish, and the ratio is undefined.

Proof. Compute:

$$\sum_{m \in \mathcal{R}} F(\lambda_m) = 14F(4) = 14(4\alpha + \beta), \quad (23)$$

$$\sum_m F(\lambda_m) = \alpha \sum_m \lambda_m + 64\beta = 256\alpha + 64\beta = 64(4\alpha + \beta). \quad (24)$$

Ratio $= 14(4\alpha + \beta)/[64(4\alpha + \beta)] = 14/64 = 7/32$, provided $4\alpha + \beta \neq 0$. If $4\alpha + \beta = 0$, then $F(4) = 0$ and both sums vanish; the ratio is undefined. $\square$

Corollary 2.23 (Geometric content). *The convention-independent content of the cancellation is*

$$\frac{\sum_{m \in \mathcal{R}} \lambda_m}{\sum_m \lambda_m} = \frac{56}{256} = \frac{7}{32}. \quad (25)$$

The numerator $14 \cdot 4 = 56$ is multiplicity times the midpoint eigenvalue; the denominator $64 \cdot 4 = 256$ is total mode count times the midpoint eigenvalue.

2.10 The TF operator and the $|\Delta n|$ grading

Definition 2.24 (Diagonal flip). The diagonal flip is $F : (n_1, n_2) \mapsto (n_2, n_1)$ on Λ_L .

Definition 2.25 (TF operator). The TF operator is the composition

$$\text{TF} = T_{\text{half}} \circ F, \quad \text{TF}(n_1, n_2) = ((n_2 + L/2) \bmod L, (n_1 + L/2) \bmod L). \quad (26)$$

Lemma 2.26. $\text{TF}^2 = \text{id}$.

Proof. $\text{TF}^2(n_1, n_2) = \text{TF}(n_2 + L/2, n_1 + L/2) = (n_1 + L, n_2 + L) = (n_1, n_2) \bmod L$. $\square$

Theorem 2.27 (TF grading by $|\Delta n|$). *Let $\Delta n = n_2 - n_1$ on Λ_8 . On the 14 modes at $\lambda = 4$:*

- *TF-fixed modes: all modes with $\Delta n \equiv 4 \pmod{8}$. There are 8 such modes: $\{\nu, e, d, b, \mu, c, Z, H\}$.*
- *TF 2-cycle modes: all modes with $\Delta n \not\equiv 4 \pmod{8}$. There are 6 such modes, forming three 2-cycles: $\{u, \tau\}$, $\{s, \text{dark}\}$, $\{W, t\}$.*

Proof. TF-fixed iff $(n_2 + 4) \bmod 8 = n_1$ and $(n_1 + 4) \bmod 8 = n_2$, equivalent to $n_2 - n_1 \equiv 4 \pmod{8}$. By the parity constraint $n_1 \equiv n_2 \pmod{2}$ (which holds on every mode at $\lambda = 4$), the condition $\Delta n \equiv 4 \pmod{8}$ is equivalent to $|\Delta n| = 4$. Direct enumeration of the 14 modes confirms the stated classification. $\square$

Corollary 2.28 (TF-fixed $\Leftrightarrow$ higher $N/4$). *On each T_{half} pair, the TF-fixed member is exactly the member with smaller n_1 , which by the HCSM-23 within-pair ordering rule receives the higher $N/4$ label.*

Remark 2.29. This corollary derives the HCSM-23 within-pair ordering rule from the TF operator of the substrate. What was stated as a rule in HCSM-23 is now a theorem of Axiom 1.

2.11 Klein four-orbit structure

Definition 2.30 (Substrate involutions). On Λ_8 , three commuting involutions act on the mode labels:

$$F : (n_1, n_2) \mapsto (n_2, n_1), \quad (27)$$

$$\text{TF} : (n_1, n_2) \mapsto ((n_2 + 4) \bmod 8, (n_1 + 4) \bmod 8), \quad (28)$$

$$F \circ \text{TF} : (n_1, n_2) \mapsto ((n_1 + 4) \bmod 8, (n_2 + 4) \bmod 8). \quad (29)$$

All three commute pairwise.

Theorem 2.31 (Klein four-orbit). *The four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful action of the Klein four-group $V_4 = \{\text{id}, F, \text{TF}, F \cdot \text{TF}\}$:*

$$F : u \leftrightarrow s, \quad \text{dark} \leftrightarrow \tau, \quad (30)$$

$$\text{TF} : u \leftrightarrow \tau, \quad s \leftrightarrow \text{dark}, \quad (31)$$

$$F \cdot \text{TF} : u \leftrightarrow \text{dark}, \quad s \leftrightarrow \tau. \quad (32)$$

The action is faithful, transitive, and free (no stabilizer).

Proof. Direct computation on the four modes. All three involutions act as products of two disjoint transpositions, pairwise commute, and generate the Klein four-group. No mode is fixed by any non-identity element, so the action is free. $\square$

Corollary 2.32. *The Klein four-orbit $\{u, s, \text{dark}, \tau\}$ is generated by the substrate involutions F and TF . This is the HCSM-27 Klein orbit, now derived from Axiom 1.*

3 Level 1: The Hodge Complex

3.1 The three form degrees

On the 2-torus Λ_L , the Hodge complex of differential forms has three graded pieces:

$$\Omega^0 = \mathbb{R}^N, \quad \Omega^1 = \mathbb{R}^{2N}, \quad \Omega^2 = \mathbb{R}^N, \quad (33)$$

with total dimension $N + 2N + N = 4N$. At $N = 64$ this gives 256.

3.2 The exterior derivative and the codifferential

Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$ be the two oriented lattice directions. The exterior derivative acts on 0-forms by

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad \mu \in \{1, 2\}, \quad (34)$$

and on 1-forms by

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x). \quad (35)$$

These satisfy $d_1 d_0 = 0$ exactly (nilpotency). The codifferential $d^* = d^\dagger$ is defined as the adjoint with respect to the standard Euclidean inner product.

3.3 The Hodge-Dirac operator

Definition 3.1 (Hodge-Dirac operator). The Hodge-Dirac operator is $D = d + d^*$, acting on the full Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$. In block form,

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}, \quad (36)$$

a symmetric 256×256 real operator.

Theorem 3.2 (Hodge-Dirac square). *On the discrete torus Λ , the identity $D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k}$ holds exactly on each form degree $k = 0, 1, 2$.*

Proof. On Ω^0 , $D^2 = d_0^* d_0 = \Delta$. On Ω^1 , $D^2 = d_0 d_0^* + d_1^* d_1 = \Delta$ by direct component expansion. On Ω^2 , $D^2 = d_1 d_1^* = \Delta$. Cross terms are removed by nilpotency $d_1 d_0 = 0$. All three identities are verified at machine precision. $\square$

Remark 3.3 (No species doubling). The Hodge-Dirac operator on the discrete torus has no species doubling: its square equals the scalar Laplacian on each form degree. The low-energy spectrum is bounded away from zero by $2 \sin(\pi/L)$ except for the four zero modes.

3.4 The $\mathbb{Z}_2$ grading and the supercharge structure

Definition 3.4 (Parity operator). The parity operator $(-1)^F$ on the Hodge complex is $+1$ on $\Omega^0 \oplus \Omega^2$ and -1 on Ω^1 . It $\mathbb{Z}_2$ -grades the complex into a bosonic sector $\Omega^0 \oplus \Omega^2$ and a fermionic sector Ω^1 .

Proposition 3.5. $D(-1)^F = -(-1)^F D$.

Theorem 3.6 (Supercharge structure). *In the $\mathbb{Z}_2$ grading, the Hodge-Dirac operator decomposes as*

$$D = Q + Q^\dagger, \quad (37)$$

where Q is the off-diagonal block of D mapping the bosonic sector to the fermionic sector, and $Q^\dagger$ is its adjoint. The supercharges satisfy

$$Q^2 = (Q^\dagger)^2 = 0, \quad \{Q, Q^\dagger\} = D^2 = \Delta, \quad (38)$$

and the Hamiltonian $H = D^2/2 = \Delta/2$ satisfies $\{Q, Q^\dagger\} = 2H$. This is the standard $N = 1$ supersymmetric quantum mechanics algebra.

Proof. In the $\mathbb{Z}_2$ grading, D is off-diagonal:

$$D = \begin{pmatrix} 0 & Q^\dagger \\ Q & 0 \end{pmatrix}, \quad (39)$$

with $Q : \Omega^0 \oplus \Omega^2 \rightarrow \Omega^1$ and $Q^\dagger : \Omega^1 \rightarrow \Omega^0 \oplus \Omega^2$. Then

$$D^2 = \begin{pmatrix} Q^\dagger Q & 0 \\ 0 & Q Q^\dagger \end{pmatrix}. \quad (40)$$

The algebra follows from nilpotency and Theorem 3.2. $\square$

3.5 The kernel and the emergent dimension

Theorem 3.7 (Kernel dimension). *The kernel of the Hodge-Dirac operator is isomorphic to the de Rham cohomology of the torus:*

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}), \quad (41)$$

with

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4. \quad (42)$$

Corollary 3.8 (Emergent dimension). $d = \dim \ker D = 4$.

Theorem 3.9 (Lorentzian signature). *The Hodge star $\star$ restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$: the ± 1 pair from the $(H^0 + H^2)$ sector, the $\pm i$ pair from the H^1 sector. The intersection form $Q(\alpha, \beta) = \int_\Lambda \alpha \wedge \beta$ on the even-degree sector and the Hodge inner product $\langle \alpha, \beta \rangle = \int_\Lambda \alpha \wedge \star \beta$ on the odd-degree sector combine into a metric*

$$g = Q|_{H^0+H^2} + \langle \cdot, \cdot \rangle|_{H^1} \quad (43)$$

of signature $(3, 1)$ on $\ker D$.

3.6 The spectral gap and absence of species doubling

Theorem 3.10 (Smallest nonzero eigenvalue of D). *On $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L even, the smallest nonzero eigenvalue of the Hodge-Dirac operator is exactly $\mu_{\min}(L) = 2 \sin(\pi/L)$.*

Corollary 3.11. *For every finite even L , the Hodge-Dirac operator has exactly four zero eigenvalues and the next eigenvalue is bounded away from zero by $2 \sin(\pi/L)$.*

3.7 The Hodge-HCSM interface: entropic flatness

The Hodge-HCSM interface is the passage from the abstract Hodge complex to the processed fermion/boson structure. The mirror-cancellation theorem of Section 2 provides the projection onto the 56-dimensional sector V_{56} , and the processor channel of this section provides the map onto the fermion sector.

3.7.1 The 56-dimensional Hodge sector

The Hodge complex at the self-paired eigenvalue $\lambda = 4$ is

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (44)$$

of total dimension $14 + 28 + 14 = 56$. Its D_4 isotypic decomposition is

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E, \quad (45)$$

with dimension check $9 + 5 + 7 + 7 + 28 = 56$.

3.7.2 The processor channel

By HCSM-16 [16] (Definition 1, Lemma 2), the block-diagonal extension map $\text{extend} : \mathbb{C}^{14 \times 14} \rightarrow \mathbb{C}^{56 \times 56}$ is a Lie algebra homomorphism. The 56-dimensional processor is

$$T_{56} = \text{extend}(T) = T \oplus (T \otimes \mathbb{I}_2) \oplus T, \quad (46)$$

with $\text{rank}(T_{56}) = 10 + 20 + 10 = 40$ and $\dim \ker(T_{56}) = 16$.

Definition 3.12 (Processor channel). The processor channel is the completely positive trace-preserving (CPTP) map

$$\Phi_T(\rho) = T_{56}\rho T_{56} + \text{Tr}((\mathbb{I}_{56} - T_{56})\rho)|0\rangle\langle 0|_{\text{bos}}, \quad (47)$$

with $|0\rangle_{\text{bos}}$ a fixed reference state on the 16-dimensional bosonic kernel.

3.7.3 Entropic flatness and the entropy reduction

Theorem 3.13 (Entropic flatness). *The natural state on the 56-dimensional Hodge sector V_{56} is the maximally mixed state*

$$\rho_{\text{in}} = \frac{\mathbb{I}_{56}}{56}, \quad (48)$$

obtained as the image of the maximally mixed state on the full Hodge complex $\rho_{\text{full}} = \mathbb{I}_{256}/256$ under the mirror-cancellation projection. Its von Neumann entropy is

$$S_{\text{in}} = -\text{Tr}(\rho_{\text{in}} \log \rho_{\text{in}}) = \log 56 \approx 4.0253516907 \text{ nats}. \quad (49)$$

Theorem 3.14 (Processor channel output). *The processor channel Φ_T maps the maximally mixed input $\rho_{\text{in}} = \mathbb{I}_{56}/56$ to the maximally mixed state on the 40-dimensional image of T_{56} , with von Neumann entropy*

$$S_{\text{out}} = \log 40 \approx 3.6888794541 \text{ nats}. \quad (50)$$

Corollary 3.15 (Entropy reduction). *The entropy reduction effected by the processor is*

$$\Delta S = S_{\text{in}} - S_{\text{out}} = \log 56 - \log 40 = \log\left(\frac{7}{5}\right) \approx 0.3364722366 \text{ nats}. \quad (51)$$

Corollary 3.16 (Framework-constant form). *The entropy reduction has the clean HCSM-native form*

$$\Delta S = \log\left(\frac{\dim V_{56}}{\text{rank}(T_{56})}\right) = \log\left(\frac{56}{40}\right) = \log\left(\frac{7}{5}\right). \quad (52)$$

Corollary 3.17 (Landauer saturation). *Conditional on the HCSM measurement-irreversibility principle (HCSM-21 [21], Theorem 12.1), the entropy reduction $\Delta S = \log(7/5)$ equals the exact deterministic Landauer bound for the processor's information-processing step.*

4 Level 2: The Spectral Midpoint

4.1 The self-paired eigenvalue

The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ of the scalar Laplacian spectrum has a unique fixed point at $\lambda = 4$. The eigenspace at $\lambda = 4$ has multiplicity $2(L - 1) = 14$ at $L = 8$. The 14 modes are

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (53)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (54)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (55)$$

4.2 The D_4 orbit decomposition

Proposition 4.1 (Orbit stabilizers). *Under the D_4 point group, the 14 modes decompose into the three orbits above of sizes $(2, 8, 4)$. The stabilizer orders of the three orbits are 4, 1, 2 respectively, satisfying the orbit-stabilizer identity $|\mathcal{O}_k| = |D_4|/|\text{Stab}(\mathcal{O}_k)| = 8/|\text{Stab}(\mathcal{O}_k)|$.*

4.3 The 56-dimensional Hodge lift

Theorem 4.2 (Fourteen-fold lift). *The 14-fold degeneracy of the scalar Laplacian at $\lambda = 4$ lifts to the Hodge complex as $14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56$.*

4.4 The D_4 decomposition of the 56-dimensional sector

Theorem 4.3 (D_4 decomposition). *As D_4 representations,*

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (56)$$

$$\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \quad (57)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (58)$$

The full 56-dimensional sector decomposes as $56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$.

5 Level 3: The Processor and Its Dynamics

5.1 The winding generator

Definition 5.1 (Winding matrix and winding generator). Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors of the 14 modes. The winding matrix is the rank-2 antisymmetric bilinear

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (59)$$

The winding generator is

$$H_{\text{wind}} = i \sin(\pi M/8), \quad (60)$$

with $\sin$ applied elementwise.

Theorem 5.2 (Characteristic polynomial). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda \mathbb{I}_{14} - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32). \quad (61)$$

Its spectrum is $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$.

5.2 The winding classes and the processor

Definition 5.3 (Winding class partition). The 14 modes partition into three winding classes:

$$A = \{\nu, e, t\}, \quad |A| = 3; \quad (62)$$

$$B_{\text{wind}} = \{u, d, s, \mu, c, \tau, b\}, \quad |B_{\text{wind}}| = 7; \quad (63)$$

$$C = \{\text{dark}, W, Z, H\}, \quad |C| = 4. \quad (64)$$

The partition is the joint eigenspace decomposition of the n_1 -parity D and the processor T (HCSM-05 [5], Theorem 7.7).

Definition 5.4 (Processor). The processor T is the rank-10 orthogonal projection $T = I - P_C$ with P_C the diagonal projector onto winding class C .

Proposition 5.5 (Fermion/boson split). *$T = I - P_C = P_A + P_{B_{\text{wind}}}$, with $\text{im}(T) = F = A \cup B_{\text{wind}}$ and $\text{ker}(T) = C = B_{\text{bos}}$. The fermion and boson sector dimensions are $|F| = d(d+1)/2 = 10$ and $|B_{\text{bos}}| = d = 4$ (HCSM-13 [13], HCSM-19 [19]). The internal ratios are $(|F| - |B_{\text{bos}}|)/|B_{\text{bos}}| = (d-1)/2 = 3/2$ and $|F|/|B_{\text{bos}}| = (d+1)/2 = 5/2$.*

5.2.1 Chirality-breaking localization

Theorem 5.6 (Chirality-breaking localization). *The commutator of the processor with the chiral reflection,*

$$C_{rs} = T \cdot P_{rs} - P_{rs} \cdot T, \quad (65)$$

is supported exactly on the two-element set $\{\text{dark}, \tau\}$. On this two-dimensional plane, the commutator is a fourth-order rotation generator:

$$C_{rs}|_{\{\text{dark}, \tau\}} = \begin{pmatrix} 0 & -1 \\ +1 & 0 \end{pmatrix}, \quad C_{rs}^2|_{\{\text{dark}, \tau\}} = -P_{\{\text{dark}, \tau\}}, \quad C_{rs}^4|_{\{\text{dark}, \tau\}} = +P_{\{\text{dark}, \tau\}}. \quad (66)$$

On the other twelve modes, T and P_{rs} commute.

Proof. Direct computation of the commutator. The processor T is diagonal with $\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1)$, and the chiral reflection P_{rs} permutes the modes by the diagonal reflection. For the commutator to be nonzero at entry (i, j) we require both $(P_{rs})_{ij} = 1$ and $T_{ii} \neq T_{jj}$. The only pair with unequal diagonal entries is $\{\text{dark}, \tau\}$ ($T_{\tau\tau} = 1$, $T_{\text{dark}, \text{dark}} = 0$). The two fixed points W and t contribute nothing. $\square$

5.2.2 The observer pair

Proposition 5.7 (Observer pair). *The chiral reflection rs fixes exactly the two modes $W = (2, 2)$ and $t = (6, 6)$. These form the observer pair of HCSM-26 [25].*

5.2.3 Klein four-orbit structure

The Klein four-orbit $\{u, s, \text{dark}, \tau\}$ of Theorem 2.31 carries the chirality-breaking pair $\{\text{dark}, \tau\}$ as a subset. The chiral reflection and the half-period translation both individually swap these two modes with other modes in the four-cycle; the product fixes them together.

5.3 The Kernel-Diagonal Theorem

Theorem 5.8 (Kernel-Diagonal). *Let $P_{\ker}$ be the orthogonal projector onto $\ker(H_{\text{wind}})$. The diagonal of $P_{\ker}$ in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\ker})_i = \begin{cases} 2/d = 1/2, & i \in \mathcal{O}_0, \\ (d-1)/d = 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (67)$$

Corollary 5.9 (Three coefficients from one diagonal). *With $P_{\ker}$ as above,*

$$c = \frac{d+1}{\text{Tr}(P_{\ker}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}, \quad (68)$$

$$\frac{\text{diag}(P_{\ker})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\ker})|_{\mathcal{O}_0}} = \frac{d-1}{2} = \frac{3}{2}, \quad (69)$$

$$\dim \ker(H_{\text{wind}}) = 1 + (d-1)^2 = 10. \quad (70)$$

Proof. Partial traces: $\text{Tr}(P_{\ker}|_{\mathcal{O}_0}) = |\mathcal{O}_0| \cdot 2/d = 2 \cdot 1/2 = 1 = d/d$ and $\text{Tr}(P_{\ker}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = |\mathcal{O}_1 \cup \mathcal{O}_2| \cdot (d-1)/d = 12 \cdot 3/4 = 9 = (d-1)^2$. Then $c = (d+1)/(d-1)^2 = 5/9$. The mass-map ratio is $(d-1)/d$ divided by $2/d$ equals $(d-1)/2 = 3/2$. Total trace is $1 + (d-1)^2 = 10 = \dim \ker(H_{\text{wind}})$. $\square$

5.3.1 Kernel- T decomposition

Theorem 5.10 (Kernel- T decomposition). *The kernel of H_{wind} decomposes under the processor T as*

$$\dim(\ker H_{\text{wind}} \cap \text{im } T) = 6, \quad \dim(\ker H_{\text{wind}} \cap \ker T) = 1. \quad (71)$$

The remaining 3 dimensions of $\ker H_{\text{wind}}$ form a mixed subspace with support on both F and C . The unique C -component of the kernel is the F -antisymmetric vector $(e_Z - e_H)/\sqrt{2}$.

Proof. Restrict H_{wind} to the coordinate subspaces F (10 columns) and C (4 columns). Let H_F and H_C denote the resulting 14×10 and 14×4 matrices. Direct computation gives $\text{rank}(H_F) = 4$ and $\text{rank}(H_C) = 3$. Then

$$\dim(\ker H_{\text{wind}} \cap \text{im } T) = 10 - \text{rank}(H_F) = 6, \quad \dim(\ker H_{\text{wind}} \cap \ker T) = 4 - \text{rank}(H_C) = 1. \quad (72)$$

The C -block H_C has rank 3. Its null direction is the vector $v_C = (e_Z - e_H)/\sqrt{2}$. The remaining 3 dimensions of the kernel lie in the mixed span $F \oplus C$. $\square$

Remark 5.11. A previous version of this result claimed $\dim(\ker H_{\text{wind}} \cap \ker T) = 4$. That claim assumed H_{wind} annihilates the bosonic class C , which is false: the block $H|_C$ has the nonzero entry $(H_{\text{wind}})_{W, \text{dark}} = i$. Theorem 5.10 corrects this.

5.4 The five-layer loop

The five-layer composition acts diagonally on the 14-mode multiplet:

$$M_{5\text{layer}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (73)$$

Theorem 5.12 (Loop-pass duration). *The loop-pass duration is selected by the deterministic Landauer bound: $T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma) = 0.7114000035169658 \dots$*

Theorem 5.13 (Exact ratio). $|M_A|/|M_B| = 3 = d - 1$.

5.4.1 Family-block eigenvalues

Theorem 5.14 (Family-block eigenvalues). *For every even L , the family blocks H_A and H_B each have rank 2, with nonzero eigenvalues exactly*

$$\text{spec}_{\neq 0}(H_A) = \text{spec}_{\neq 0}(H_B) = \{+L/2, -L/2\}. \quad (74)$$

5.4.2 Four kernel constants

Theorem 5.15 (Kernel constants at $L = 8$). *At $L = 8$, the kernel of $H_{\text{wind}}(8)$ contains the four parity-block indicators*

$$\mathbf{1}_{A_{\text{odd}}}, \quad \mathbf{1}_{A_{\text{even}}}, \quad \mathbf{1}_{B_{\text{odd}}}, \quad \mathbf{1}_{B_{\text{even}}}. \quad (75)$$

Their supports are pairwise disjoint, so they span a 4-dimensional subspace of $\ker H_{\text{wind}}(8)$. The fifth indicator $\mathbf{1}_{AB}$ is not in the kernel: $\|H_{\text{wind}}(8)\mathbf{1}_{AB}\| = 4$.

5.4.3 Rank formula

Theorem 5.16 (Rank formula). *For every even $L \geq 4$,*

$$\text{rank} H_{\text{wind}}(L) = \begin{cases} L, & L \equiv 2 \pmod{4}, \\ 4\lceil L/8 \rceil, & L \equiv 0 \pmod{4}. \end{cases} \quad (76)$$

Corollary 5.17 (Nullity). *The nullity of $H_{\text{wind}}(L)$ is $2(L - 1) - \text{rank} H_{\text{wind}}(L)$.*

5.5 The loop operator and its commutant

This section integrates the structural results of HCSM-15 [15]. The loop operator M_{loop} is the diagonal class operator of HCSM-17 [17] Definition 2.1, distinct from the five-layer composition $M_{5\text{layer}}$.

Definition 5.18 (Loop operator). The loop operator on the 14-mode multiplet is the diagonal operator

$$M_{\text{loop}} = M_A P_A + M_B P_B + M_C P_C, \quad (77)$$

with M_A, M_B, M_C the complex loop eigenvalues of HCSM-14 [14] and HCSM-17 [17], and $|M_A|/|M_B| = 3$ exactly. In this paper we take $(M_A, M_B, M_C) = (1, 2, 0)$ without loss of generality: any other choice of distinct eigenvalues gives the same loop commutant.

M_{loop} is not Hermitian (M_A, M_B are complex), but it is normal: $M_{\text{loop}} M_{\text{loop}}^\dagger = M_{\text{loop}}^\dagger M_{\text{loop}}$ because it is diagonal. It commutes with each class projector:

$$[M_{\text{loop}}, P_A] = [M_{\text{loop}}, P_B] = [M_{\text{loop}}, P_C] = 0. \quad (78)$$

Definition 5.19 (Loop commutant). The loop commutant is the intersection

$$\mathfrak{g}_{\text{loop}} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}}) \subseteq \mathfrak{g}_{\text{SM}}. \quad (79)$$

Theorem 5.20 (Loop commutant dimension). *The loop commutant $\mathfrak{g}_{\text{loop}}$ is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$.*

Proof. Form the stacked commutator matrix $C \in \mathbb{C}^{(14)^2 \times 16}$ whose j -th column is $\text{vec}([M_{\text{loop}}, g_j])$ for $g_j \in \mathfrak{g}_{\text{SM}}^{(14)}$. The realified matrix

$$C_{\mathbb{R}} = \begin{pmatrix} \text{Re}(C) \\ \text{Im}(C) \end{pmatrix} \in \mathbb{R}^{2(14)^2 \times 16} \quad (80)$$

has rank 13 at machine precision, with a singular value gap of $\sim 10^{15}$ between the 13th and 14th singular values. The nullspace dimension is $16 - 13 = 3$.

The three nullspace vectors $\{v^{(1)}, v^{(2)}, v^{(3)}\} \subset \mathbb{R}^{16}$ span a real 3-dimensional subspace $N \subseteq \mathbb{R}^{16}$. Each vector generates a Hermitian operator

$$O_i = \sum_{j=1}^{16} v_j^{(i)} g_j \in \mathfrak{g}_{\text{SM}} \quad (81)$$

satisfying $\|[O_i, M_{\text{loop}}]\| < 10^{-15}$ and $\|O_i^\dagger - O_i\| = 0$. Since the g_j are Hermitian and the $v_j^{(i)}$ are real, every O_i is Hermitian; the span of the O_i is therefore a real 3-dimensional Hermitian subspace of $\mathfrak{g}_{\text{SM}}$.

Cross-commutators $[O_i, O_j]$ vanish at machine precision ($\|[O_i, O_j]\| < 10^{-15}$), so the algebra is abelian. Since each O_i is a real linear combination of the 16 Hermitian generators of $\mathfrak{g}_{\text{SM}}$, and the commutator of any two such combinations can be expanded in the structure constants of $\mathfrak{g}_{\text{SM}}$, direct computation gives zero at machine precision. $\square$

Remark 5.21. The loop commutant $\mathfrak{g}_{\text{loop}}$ is a proper 3-dimensional subalgebra of the 16-dimensional $\mathfrak{g}_{\text{SM}}$. The remaining 13 directions of $\mathfrak{g}_{\text{SM}}$ are broken by the loop operator: they correspond to generators that do not commute with M_{loop} .

Theorem 5.22 (Isotypic projector degeneracy). *Every element of the loop commutant has equal coefficients on the isotypic projectors P_{A_2} , P_{B_2} , and P_E .*

Proof. Let $v \in \mathbb{R}^{16}$ be any nullspace vector. The components of v corresponding to the three isotypic projectors P_{A_2}, P_{B_2}, P_E are the entries v_{A_2}, v_{B_2}, v_E . Direct computation gives

$$|v_{A_2} - v_{B_2}| < 10^{-15}, \quad |v_{A_2} - v_E| < 10^{-15}, \quad |v_{B_2} - v_E| < 10^{-15}. \quad (82)$$

The class partition of HCSM-14 [14] is blind to the distinction between the A_2 , B_2 , and E isotypic sectors. $\square$

Theorem 5.23 (No root generators). *The loop commutant has zero overlap with the root generators $\lambda^1, \lambda^2, \lambda^4, \lambda^5, \lambda^6, \lambda^7$ of $\mathfrak{su}(3)_c$ and σ^1, σ^2 of $\mathfrak{su}(2)_L$. Only the Cartan generators $\lambda^3, \lambda^8, \sigma^3$ of $\mathfrak{g}_{\text{SM}}$ have nonzero overlap with the loop commutant.*

Proof. For each root generator g_j , project the unit coefficient vector $e_j \in \mathbb{R}^{16}$ onto the nullspace N via the orthogonal projector P_N :

$$\|P_N e_j\| < 10^{-15}, \quad j \in \{1, 2, 4, 5, 6, 7, 9, 10\}. \quad (83)$$

The vanishing residuals reflect the fact that the loop operator breaks all off-diagonal (root) directions of $\mathfrak{su}(3)_c$ and $\mathfrak{su}(2)_L$. Only the diagonal (Cartan) generators $\lambda^3, \lambda^8, \sigma^3$ have nonzero overlap with $\mathfrak{g}_{\text{loop}}$. $\square$

Corollary 5.24 (Loop commutant basis). *The loop commutant $\mathfrak{g}_{\text{loop}}$ is spanned by three Hermitian combinations of $\{\lambda^3, \lambda^8, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}$, in which the coefficients on P_{A_2}, P_{B_2}, P_E are equal.*

5.6 The DNLS Hartree Hamiltonian and its commutant

Definition 5.25 (DNLS Hartree kernel). For each mode i of the 14-mode multiplet, let $\phi_i(x, y)$ denote the normalized real wavefunction on the torus. The Hartree kernel is

$$K_{ab} = \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad a, b = 1, \dots, 14. \quad (84)$$

The diagonal of K equals the inverse participation ratio IPR_a , taking values $(2/128, 3/128, 4/128)$ on the three D_4 orbits.

Definition 5.26 (DNLS Hartree Hamiltonian). The DNLS Hartree Hamiltonian on the 14-mode multiplet is

$$H_{\text{DNLS}} = 4\mathbb{I}_{14} + UK, \quad (85)$$

with $U = B_{\text{mass}} = (d-1)kL^2/(2N^2) = 0.5411967342\dots$ the mass-map rate (HCSM-19 [19], HCSM-33 [29]).

H_{DNLS} is a real symmetric 14×14 matrix, hence Hermitian. It commutes with no particular class projector in general, but as we prove below, it commutes with the full $\mathfrak{su}(2)_L$ factor of $\mathfrak{g}_{\text{SM}}$.

Definition 5.27 (DNLS commutant). The DNLS commutant is

$$\mathfrak{g}_{\text{DNLS}} = \mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}). \quad (86)$$

Theorem 5.28 (DNLS commutant dimension). *The DNLS commutant $\mathfrak{g}_{\text{DNLS}}$ is a ten-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$.*

Proof. The realified DNLS commutator matrix has rank 6 and nullity 10 at machine precision, with a singular value gap of $\sim 10^{13}$ between the 6th and 7th singular values.

The nullspace decomposes explicitly as

$$\mathfrak{g}_{\text{DNLS}} = \mathfrak{su}(2)_L \oplus \langle P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E \rangle \oplus \mathfrak{c}, \quad (87)$$

where $\mathfrak{c}$ is a two-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$. Dimension count: $3 + 5 + 2 = 10$.

The $\mathfrak{su}(2)_L$ generators $\sigma^1, \sigma^2, \sigma^3$ are non-abelian, with structure constants $[\sigma^i, \sigma^j] = i\epsilon^{ijk}\sigma^k$ matching those of $\mathfrak{su}(2)$. The two $\mathfrak{c}$ generators are both Hermitian, traceless on the A_1 block, and commute with one another ($[O_i, O_j] = 0$ at machine precision); they are therefore a 2-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$. $\square$

5.6.1 Doubled-momentum pairing

Lemma 5.29 (Doubled-momentum pairing). *Let $U_{B_1} = [v_1^{(B_1)}, v_2^{(B_1)}] \in \mathbb{R}^{14 \times 2}$ be the explicit orthonormal basis of the B_1 isotypic sector constructed in Appendix A. The two basis vectors pair modes of the multiplet such that for each pair $\{m, m'\}$ in the support of $v_1^{(B_1)}$ and $v_2^{(B_1)}$, the doubled momenta agree:*

$$(2n_1^{(m)}, 2n_2^{(m)}) \equiv (2n_1^{(m')}, 2n_2^{(m')}) \pmod{8}, \quad (88)$$

and the mode densities satisfy

$$|\phi_m(x)|^2 = |\phi_{m'}(x)|^2, \quad (89)$$

with the sign opposite between the two B_1 basis vectors.

Proof. See Appendix A. The B_1 isotypic sector is spanned by the two-dimensional vector representation of D_4 . The explicit basis vectors constructed in Appendix A pair the modes as follows: v_1 pairs $\nu = (0, 4)$ and $e = (4, 0)$, both with doubled momentum $(0, 0) \pmod{8}$; v_2 pairs $u = (1, 3)$ with $\text{dark} = (5, 7)$, $\tau = (7, 5)$ with $s = (3, 1)$, $d = (1, 5)$ with $b = (5, 1)$, and $c = (7, 3)$ with $\mu = (3, 7)$. In each pair, the two modes have the same doubled momentum. Because K depends only on the squares $|\phi_a|^2 |\phi_b|^2$, the off-diagonal entries of K_{B_1} are sums of terms with equal magnitudes and opposite signs. $\square$

Theorem 5.30 (Doubled-momentum pairing theorem). *The B_1 -compressed Hartree kernel vanishes identically:*

$$K_{B_1} = U_{B_1}^T K U_{B_1} = 0 \in \mathbb{R}^{2 \times 2}. \quad (90)$$

Proof. The compression K_{B_1} has entries

$$(K_{B_1})_{ij} = (v_i^{(B_1)})^T K v_j^{(B_1)} = \sum_{a,b} v_i^{(a)} K_{ab} v_j^{(b)}, \quad i, j \in \{1, 2\}. \quad (91)$$

Substituting $K_{ab} = \sum_x |\phi_a(x)|^2 |\phi_b(x)|^2$:

$$(K_{B_1})_{ij} = \sum_x \left[\sum_a v_i^{(a)} |\phi_a(x)|^2 \right] \left[\sum_b v_j^{(b)} |\phi_b(x)|^2 \right]. \quad (92)$$

Define the two density profiles

$$\rho_i(x) = \sum_a v_i^{(a)} |\phi_a(x)|^2, \quad i = 1, 2. \quad (93)$$

Then $(K_{B_1})_{ij} = \sum_x \rho_i(x) \rho_j(x)$. We must show $\sum_x \rho_1(x) \rho_2(x) = 0$ and $\sum_x \rho_1(x)^2 = \sum_x \rho_2(x)^2 = 0$.

By Lemma 5.29, the two basis vectors $v_1^{(B_1)}$ and $v_2^{(B_1)}$ pair modes with the same doubled momentum and opposite signs. For each pair $\{m, m'\}$ in the support, the contributions to $\rho_1(x)$ and $\rho_2(x)$ are

$$\rho_1(x) \supset v_1^{(m)} |\phi_m(x)|^2 + v_1^{(m')} |\phi_{m'}(x)|^2 = c_1 (|\phi_m(x)|^2 - |\phi_{m'}(x)|^2) = 0, \quad (94)$$

using $|\phi_m|^2 = |\phi_{m'}|^2$ from Lemma 5.29. Similarly $\rho_2(x) = 0$. Hence $\rho_1 = \rho_2 = 0$ identically, and $K_{B_1} = 0$. $\square$

Corollary 5.31 ($\mathfrak{su}(2)_L$ preservation). *The three generators $\sigma^1, \sigma^2, \sigma^3$ commute with the DNLS Hartree Hamiltonian:*

$$[\sigma^i, K] = 0, \quad [\sigma^i, H_{\text{DNLS}}] = 0, \quad i = 1, 2, 3. \quad (95)$$

Each of $\sigma^1, \sigma^2, \sigma^3$ has overlap 1 with the DNLS commutant at machine precision.

Proof. Since $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ has support only on the B_1 isotypic sector, we have

$$[T_L^i, K] = U_{B_1} [\sigma^i, K_{B_1}] U_{B_1}^T + (\text{terms acting on non-}B_1 \text{ sectors that vanish by support}). \quad (96)$$

By Theorem 5.30, $K_{B_1} = 0$, so $[\sigma^i, K_{B_1}] = 0$ identically. Hence $[T_L^i, K] = 0$ for all i . Since $[\sigma^i, \mathbb{I}] = 0$ trivially, $[T_L^i, H_{\text{DNLS}}] = 0$. The overlap of σ^i with the DNLS commutant nullspace is thus 1 by direct computation. $\square$

5.6.2 $\mathfrak{su}(3)_c$ breaking

Theorem 5.32 ($\mathfrak{su}(3)_c$ breaking). *The DNLS commutant contains a two-dimensional abelian subalgebra $\mathfrak{c}$ of $\mathfrak{su}(3)_c$. The root generators $\lambda^2, \lambda^5, \lambda^7$ have zero overlap with the DNLS commutant; the Cartan generators λ^3, λ^8 have partial overlap.*

Proof. Project each $\mathfrak{su}(3)_c$ generator onto the DNLS nullspace $N_{\text{DNLS}} \subseteq \mathbb{R}^{16}$ via the orthogonal projector $P_{N_{\text{DNLS}}}$. Direct computation gives

$$\|P_{N_{\text{DNLS}}} e_j\| < 10^{-10} \quad \text{for } j \in \{2, 5, 7\}, \quad (97)$$

so $\lambda^2, \lambda^5, \lambda^7$ are orthogonal to $\mathfrak{g}_{\text{DNLS}}$. The remaining five $\mathfrak{su}(3)_c$ generators $\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8$ have partial overlap with N_{DNLS} ; the two-dimensional abelian subalgebra $\mathfrak{c}$ is a specific Hermitian combination of these five generators that commutes with K . Because $\mathfrak{c}$ has dimension 2 and is abelian, it is a Cartan-like subalgebra of $\mathfrak{su}(3)_c$. $\square$

Remark 5.33. The breaking pattern $\mathfrak{su}(3)_c \rightarrow \mathfrak{su}(3)_c^{\text{Cartan}}$ (2-dimensional abelian) is the color analog of the electroweak symmetry breaking pattern, but it occurs at the substrate level (the mean-field Hartree dynamics) rather than at the electroweak scale. It reflects the mismatch between the A_1 isotypic sector's mode-momentum pairing and the B_1 isotypic sector's doubled-momentum pairing.

Corollary 5.34 (Contrast with the loop commutant). *The DNLS commutant (10-dimensional, $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$) is strictly larger than the loop commutant (3-dimensional, abelian). The two commutants share the $\mathfrak{u}(1)^5$ isotypic projector sector but differ on the $\mathfrak{su}(3)_c$ and $\mathfrak{su}(2)_L$ content: the loop operator preserves only the Cartan generators $\lambda^3, \lambda^8, \sigma^3$ (three abelian directions), while the DNLS Hamiltonian preserves the full $\mathfrak{su}(2)_L$ factor and a two-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$.*

5.7 The zero state

The zero state s is the unique nullspace vector of a 14×19 measurement matrix M of vacuum-native observables. The matrix has rank 13, nullity 1.

Theorem 5.35 (Zero-state structure). *The zero state satisfies $s(rs(m)) = -s(m)$ for all modes m , where rs is the diagonal reflection $(n_1, n_2) \mapsto (n_2, n_1)$. The two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have $s(W) = s(t) = 0$ exactly. The six independent amplitudes satisfy $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$. Weight fractions: 3.3% on $\mathcal{O}_0$, 86.5% on $\mathcal{O}_1$, 10.2% on $\mathcal{O}_2$.*

Remark 5.36 (Zero state as sort key). The zero state is not only the unique nullvector of the measurement matrix; it is also the sort key of the charge assignment and one of the three native operators used in the Midpoint Sorting Theorem (HCSM-23 Rev. 7 [23]).

5.8 The 5 + 9 split

The 5 + 9 split of the 14-mode multiplet (HCSM-11 [11]) refines the processor's action. The classification is defined by the residual

$$\delta = N/4 - \text{base}, \quad (98)$$

where base is a reference depth in the entropy well.

Definition 5.37 (Substrate Eigenmode and Mesh Resonance). A mode is a Substrate Eigenmode if $\delta = 0$; it is a Mesh Resonance if $\delta \neq 0$.

Theorem 5.38 (Derived base formula). *For each mode at $\lambda = 4$, let (k, node) be its T_{half} pair label. The residual $\delta = N/4 - \text{base}$ is given by*

$$\delta_{\text{high}} = (d-1)(2-k) + (d-1)^3 \cdot \mathcal{K}[\text{node} = \text{AB}], \quad (99)$$

$$\delta_{\text{low}} = \text{frac}(N/4_{\text{low}}) + 2(d-1) \cdot \mathcal{K}[\text{node} = \text{AB}], \quad (100)$$

where $\text{frac}(x) = x - \lfloor x \rfloor$ and $\mathcal{K}[\cdot]$ is the indicator function. The base is $\text{base} = N/4 - \delta$.

Theorem 5.39 (5 + 9 split). *Of the 14 modes at $\lambda = 4$, exactly 5 are Substrate Eigenmodes and exactly 9 are Mesh Resonances. The 5 Substrate Eigenmodes are the dark photon, the τ lepton, the W boson, the Z boson, and the Higgs boson. The 9 Mesh Resonances are the remaining 9 fermions.*

Theorem 5.40 (Shell formula). *The shell of each mode is determined by the $N/4$ value through the step function*

$$\text{shell}(N/4) = \begin{cases} 0, & N/4 \geq 50, \\ 1, & 18 \leq N/4 < 50, \\ 1.5, & 12 \leq N/4 < 18, \\ 2, & 6 \leq N/4 < 12, \\ 3, & N/4 < 6. \end{cases} \quad (101)$$

The thresholds are framework constants: $\{2(d-1), d(d-1), 2(d-1)^2, N-2(L-1)\} = \{6, 12, 18, 50\}$.

Theorem 5.41 (W/Z base formula). *The base values of the W and Z bosons are $\text{base}(W) = \text{base}(Z) = d/2 = 2$.*

5.9 The four-layer architecture

The 14-mode multiplet carries four independent structural layers (HCSM-11 [11], Theorem 9.1).

Theorem 5.42 (Layer independence). *The 14-mode multiplet carries four independent structural layers:*

1. **Winding layer:** the kernel projector P_{ker} of H_{wind} , producing $c = 5/9$, $(d-1)/2 = 3/2$, and kernel dimension 10 (HCSM-14.a).
2. **Hartree layer:** the mode-density kernel K , producing the eigenvalue $7/32$ and the coefficient $26 = 2^{d+1} - (|F| - |B_{\text{bos}}|)$.
3. **Sorting layer:** the T_{half} permutation operator P_T , producing the base formula, the 5 + 9 split, the shell values, and the W/Z base $d/2$.
4. **Processing layer:** the shape parameters r_a and the coefficients $\{7.88, 4.23\}$ of $m_f(r)$, which are orthogonal to layers 1, 2, and 3.

Layers 1, 2, and 3 are HCSM-native and mutually compatible: $[P_T, K] = 0$, $[P_{\text{ker}}, F] = 0$, $[P_{\text{ker}}, D] = 0$, and K commutes with F and D . Layer 4 is orthogonal to all three.

Remark 5.43 (Structural meaning). Layer 4 is the framework’s empirical anchor in the mass sector. It is structurally parallel to the Standard Model’s Yukawa couplings: a set of external calibration values that the framework requires but does not derive. The framework’s strength is that layers 1, 2, and 3 produce the base formula, the 5 + 9 split, the shell values, and the mirror residual with zero empirical anchors.

5.10 The mirror residual as a Hartree eigenvalue

Theorem 5.44 (Mirror residual equals Hartree eigenvalue). *The top eigenvalue of the Hartree kernel K equals the mirror-cancellation residual of Section 2:*

$$\lambda_{\max}(K) = \frac{7}{32} = \frac{2d-1}{2^{d+1}}. \quad (102)$$

The corresponding eigenvector is the uniform superposition of all 14 modes.

Proof. Direct computation. The eigenvector is $\mathbf{u} = \frac{1}{\sqrt{14}}(1, 1, \dots, 1)^T$. Applying K to $\mathbf{u}$:

$$(K\mathbf{u})_a = \sum_b K_{ab} \cdot \frac{1}{\sqrt{14}} = \frac{1}{\sqrt{14}} \sum_b \sum_x |\phi_a(x)|^2 |\phi_b(x)|^2. \quad (103)$$

By the completeness relation $\sum_b |\phi_b(x)|^2 = 1/2$ (averaged over the 14-mode subspace), the sum evaluates to $7/32$. $\square$

Theorem 5.45 (T_{half} commutes with the Hartree kernel). *The T_{half} permutation operator P_T and the Hartree kernel K commute: $[P_T, K] = 0$.*

Proof. Direct numerical verification: $\|[P_T, K]\|_F = 8.5 \times 10^{-17}$, machine zero. $\square$

Theorem 5.46 (Coefficient 26). *The HCSM-30 processing function coefficient 26 satisfies*

$$26 = 2^{d+1} - (|F| - |B_{\text{bos}}|) = 32 - 6. \quad (104)$$

6 Level 4: Output Structures

6.1 The charge functional

The electric charge in units of $e/3$ is given by the functional

$$q_3 = F(n_1, \sigma), \quad \sigma = \text{sign}(s), \quad (105)$$

where n_1 is the mode’s first momentum index and s is the zero-state amplitude.

Theorem 6.1 (Charge bijection). *The 14 modes have 14 pairwise distinct (n_1, σ) tuples. The map F is well-defined on the realized pairs and is a bijection onto the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$.*

Theorem 6.2 (σ -line sums). *The charge sums over the three σ -lines are*

$$\sum_{\sigma=-1} q_3 = 0, \quad \sum_{\sigma=0} q_3 = +5, \quad \sum_{\sigma=+1} q_3 = -8. \quad (106)$$

The total vacuum charge is $\sum_i q_{3i} = -3 = -1e$.

6.2 The rigidity class of the charge assignment

Theorem 6.3 (Charge rigidity class). *Among the 5^{14} assignments of charge values $\{-3, -1, 0, +2, +3\}$ to the 14 modes at $\lambda = 4$, exactly two assignments simultaneously satisfy the following five structural constraints:*

- (C1) *SM multiset: the multiset of charges is $\{-3, -1, 3, 0, 4, +2, 3, +3, 1\}$.*
- (C2) *σ -line sums: the charge sums over the three zero-state sign lines are 0, $d + 1 = 5$, and $-2d = -8$.*
- (C3) *F-pair sum rules: within each $|\Delta n|$ shell, the pair charge sum is rank- d (for $|\Delta n| = 4$) or $(d + 1) - d$ rank (for $|\Delta n| = 2$).*
- (C4) *$n_1 + n_2 = 4$ line: the five modes on the line $n_1 + n_2 = 4$ carry the five charge values exactly once each.*
- (C5) *Processor bookkeeping: the image-of- T charge sum is -2 and the kernel-of- T charge sum is $+1$.*

Proof. Exhaustive enumeration. The F -pair sum rules (C3) reduce the assignment space to 192 configurations. Constraints (C1), (C2), and (C4) reduce this to 3; constraint (C5) reduces this to exactly 2. $\square$

Theorem 6.4 (Charge assignment solutions). *The two rigid solutions are*

$$Q_3^{(1)} = (-3, 0, -1, -1, 2, -3, 0, 2, -3, -1, 3, 0, 0, 2), \quad (107)$$

$$Q_3^{(2)} = (0, -3, 2, -1, -1, -3, 0, 2, -3, -1, 3, 0, 0, 2). \quad (108)$$

The two solutions differ only by the linked swap on the pair $\{\nu, e\}$ and the pair $\{u, s\}$.

Theorem 6.5 (Total charge rigidity). *Both solutions have total charge $\sum_i q_{3i} = -3 = -1e$. Consequently, every assignment satisfying the five structural constraints has total charge $-1e$:*

$$\sum_i q_{3i} = -(d - 1) = -1e. \quad (109)$$

The total vacuum charge is a rigidity-class invariant.

Corollary 6.6 (Empirical residue of the charge sector). *The charge assignment is determined up to exactly one binary choice: which member of the pair $\{\nu, e\}$ carries charge 0 and which carries -3 . The Standard Model content resolves this choice: the electron e carries $-1e$ and the neutrino ν carries 0. This binary choice is now derived, not disclosed: HCSM-23 Rev. 7 [23] Corollary 4.4 shows that the within-pair ordering rule of the Midpoint Sorting Theorem forces $\nu = (0, 4)$ and $e = (4, 0)$.*

Remark 6.7 (Physical reading: what the machine must not cancel). The rigidity theorem states that the machine cannot cancel its residual charge. Any attempt to modify the charge assignment away from the two-element family violates one of the five structural constraints. The total charge $-1e$ is not a choice made by the machine; it is the unique value permitted by the machine's own structure.

6.3 The mass map

The mass map is

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i), \quad (110)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224 \text{ GeV}, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.5411967342 \dots \quad (111)$$

The map reproduces the twelve charged fermion masses and the W , Z , H , and t masses at RMS 0.0449 dex across thirteen orders of magnitude. The coefficient $(d-1)/2$ in B_{mass} is derived in HCSM-19 [19] by two independent routes (the processor F/B split and the DNLS Hartree orbit trace). The mass map itself is derived in HCSM-33 [29].

6.4 The Midpoint Sorting Theorem (summary)

The identification of the 14 modes with one Standard Model generation is now a theorem (HCSM-23 Rev. 7 [23]). The sorting uses the T_{half} involution $(n_1, n_2) \mapsto (n_1 + 4, n_2 + 4)$, the pair index $k = \min(n_1)$, the parity-block node from the sign of the signed coordinate v of the smaller- n_1 member, the (S, D) table of HCSM-23 Rev. 6, and the within-pair ordering rule.

7 Level 5: Emergent Physics

7.1 Gauge algebra from the D_4 commutant

The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ is derived as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet (HCSM-06 [6]). The A_1 isotypic block is uniquely identified with physical color by a dimension-and-invariance theorem (HCSM-06 [6], Theorem 5.4). The B_1 block is identified with physical weak isospin up to the outer-automorphism labeling convention (HCSM-06 [6], Theorem 5.10).

7.2 Hypercharge quantization

The hypercharge functional is a specific element of the 15-dimensional sector-coarse space (HCSM-06 [6], Theorem 6.3; HCSM-07 [7]). The A_1 coefficients are $(0, -1/6, -1/2)$ and the B_1 coefficients are $(0, +1/2, +1/2)$, uniquely determined by the Standard Model content and the sector assignment (HCSM-07 [7], Theorems 5.1 and 6.1). The hypercharge values are derived from the content structure and the anomaly cancellation conditions (HCSM-07 [7], Theorem 8.1). The content structure is derived from the 14-mode charges, the D_4 isotypic multiplicities, and the Hodge-lift chirality structure (HCSM-07 [7], Theorem 8.3). All values lie in $(1/6)\mathbb{Z}$.

7.3 Flavor: the CKM matrix

The Cabibbo angle satisfies $\sin \theta_{12} = \pi/14 = 0.2243995 \dots$, matching observation to 0.044% (HCSM-34 [30], Theorem 1). The winding amplitude $a = 3/5$ is topological: $a = (|F| - |B_{\text{bos}}|)/|F| = 6/10$. The Wolfenstein parameters are $\lambda = \pi/14$, $A = (d/(d+1))(1 + \delta_{\text{edge}})$, and $\sqrt{\rho^2 + \eta^2} = (d+1)/(d(d-1)) = 5/12$ (HCSM-34 [30], Theorem 3). The full CKM hierarchy is derived in HCSM-34 [30], Theorems 5 and 9.

7.4 Flavor: the PMNS matrix

The neutrino $N/4$ spectrum is $(60, 60, 58)$, derived from the four spin structures of the torus with Dirac indices $\pm 1/4$ (HCSM-36 [31], Theorem 4.1). The PMNS magnitudes match at RMS 0.035 (HCSM-36 [31], Theorem 6.1). The mixing angles are $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$;

the Dirac CP phase is $\delta_{CP} \approx 197^\circ$. The generation phase $\phi_{\text{gen}} = (3/5)\pi$ is exact (HCSM-36 [31], Section 5).

7.5 Higgs sector

The top-Yukawa coefficient is $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$ with $4/3 = |\mathcal{O}_2|/(d-1)$ (HCSM-40 [32]). The top mass is $m_t = 172.688398$ GeV (PDG 172.690 ± 0.30 GeV). The tree-level quartic coupling is $\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2]$, and the electroweak vacuum expectation value is $v_{\text{EW}} = M_P \exp(-kL_{\text{phys}})$.

7.6 Gravity

The graviton is the self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, consisting of $54B_1 \oplus 54B_2 = 108$ modes (HCSM-45 [37], HCSM-46 [38]). It is exactly massless at all orders by D_4 Schur orthogonality (HCSM-46 [38]). The Newton constant is $G_N = (d+1)^2/(8\pi M_P^2) = 25/(8\pi M_P^2)$ (HCSM-48 [39]).

7.7 Absence of a light dark photon

Theorem 7.1 (Absence of a light dark photon). *The kinetic mixing parameter satisfies $\epsilon \equiv 0$ exactly over all mass scales.*

7.8 Spin chirality $\mathbb{Z}_2$ grading

The 56-dimensional Hodge lift at $\lambda = 4$ carries three commuting $\mathbb{Z}_2$ operators: $\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$ (Hodge form-parity), χ_{spin} (spin-structure parity), and $F : (n_1, n_2) \mapsto (n_2, n_1)$ (orientation flip). All three square to I and pairwise commute, generating $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ (HCSM-42 [34]).

7.9 Strong CP from orientation $\mathbb{Z}_2$

Theorem 7.2 (Strong CP). $\theta_{\text{QCD}} = \theta_{\text{eff}} = 0$ exactly.

Proof. On the 14-mode multiplet at $\lambda = 4$, $[F, \Delta] = 0$ and $\{F, H_{\text{wind}}\} = 0$ exactly. The winding matrix M is antisymmetric under F ; the odd function $\sin(\pi M/8)$ preserves the antisymmetry. The physical identification $H_{\text{wind}} \leftrightarrow F_{\mu\nu}F^{\mu\nu}$ gives $\theta_{\text{eff}} = 0$ exactly (HCSM-43 [35]). $\square$

7.10 Proca sector and $\mathfrak{u}(1)_Y$ origin

The $\mathfrak{u}(1)_Y$ gauge symmetry is generated by the hypercharge functional $\hat{Y}$ on $\Omega^0|_{\lambda=4}$, with $[\hat{Y}, T] = 0$, $[\hat{Y}, F] = 0$, and eigenvalues in $(1/6)\mathbb{Z}$. The Proca Lagrangian is

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}W_{\mu\nu}^a W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + \frac{1}{2}m_W^2 W_\mu^+ W^{-\mu} + \frac{1}{2}m_Z^2 Z_\mu Z^\mu, \quad (112)$$

with $m_W = gv_{\text{EW}}/2$ and $m_Z = \sqrt{g^2 + g'^2}v_{\text{EW}}/2$. The mass ratio $m_W/m_Z = \cos\theta_W = 0.8769$ matches PDG 0.8814 at 0.5%. Tree-level $\rho = 1$ exactly (HCSM-44 [36]).

8 Level 6: Cosmology

8.1 The kL crossover

The bare warp factor is exact: $kL_{\text{bare}} = (d-1)N/(d+1) = 3 \cdot 64/5 = 192/5 = 38.4$. The physical warp factor is $kL_{\text{phys}} = 38.442527$, matching $\ln(M_P/v_{\text{EW}})$ to 0.004%. The correction is $kL_{\text{phys}} - kL_{\text{bare}} = ((|D_4| - 1)/2)\delta_{\text{edge}} = (7/2)\delta_{\text{edge}}$ (HCSM-51 [40]).

8.2 Dark energy and the DME entropy-well normalization

The bare dark energy density is

$$\rho_{DE}^{\text{bare}} = \frac{|F|}{|B_{\text{bos}}|} m_\nu^4 = \frac{d+1}{2} m_\nu^4 = 2.4809 \times 10^{-47} \text{ GeV}^4, \quad (113)$$

matching observation to 0.76%. The refined formula is

$$\rho_{DE} = \frac{d+1}{2} (1 + c\delta) m_\nu^4, \quad (114)$$

with $c = (d+1)/(d-1)^2 = 5/9$ and $\delta = (1-\gamma)^2/(2\pi)$, giving $\rho_{DE} = 2.5003 \times 10^{-47} \text{ GeV}^4$, matching observation to 0.010% (HCSM-52 [41], HCSM-53 [42]).

The DME entropy-well normalization is

$$S_0 = v_{\text{EW}}^2 \cdot \frac{7}{40} \cdot \left(1 + \frac{\delta_{\text{edge}}}{6}\right) = 10630.722 \text{ GeV}^2, \quad (115)$$

matching the implied value at relative residual 2.44×10^{-6} (HCSM-53 [42]).

8.3 Vacuum stability

The stability coefficient is

$$\gamma = \frac{c_2}{I(2)} = 0.702839467990 \dots \quad (116)$$

with $I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2})$, $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$, and $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$. The stability theorem states $S''(0)/S_0 = -2\gamma < 0$: the equilibrium $D = 4$ is a strict local maximum of the vacuum entropy (HCSM-54 [43]).

8.4 The Λ -promotion mechanism

The residual vacuum energy after mirror cancellation is promoted to the observed cosmological constant by pure projection, not by gravitational coupling.

The five-layer composition M_{5layer} acts on the 14-mode multiplet. The EFT layer L_{EFT} acts as a soft projector with weight $1 + c\delta$, where $c = 5/9$ is set by the Kernel-Diagonal Theorem. The residual energy projects onto the timelike direction of $\ker D$, with unit projection weight. The hidden $u(1)$ factors contribute zero.

The mechanism is therefore a projection, not a coupling. There is no gravitational suppression factor $1/M_P^2$; the residual vacuum energy is already at the right scale because it is topological. Combined with the absence of a light dark photon, this confirms that no gauge mediator exists between the hidden and visible sectors (HCSM-56.a [45]).

9 Falsifiable Predictions

1. **No light dark photon.** The kinetic mixing parameter satisfies $\epsilon \equiv 0$ exactly. Falsifiable by SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, and HL-LHC at $\epsilon > 10^{-12}$.
2. **Strong CP.** $\theta_{\text{QCD}} = 0$ exactly. Any non-zero measurement of the neutron electric dipole moment at the level predicted by a generic θ_{QCD} would falsify the framework.
3. **ρ parameter.** Tree-level $\rho = m_W^2/(m_Z^2 \cos^2 \theta_W) = 1$ exactly. Any measured deviation of ρ from 1 at tree level would falsify the framework.

4. **Fermion content.** One Standard Model generation contains exactly 14 particles. A fourth generation would falsify the framework.
5. **Kernel structure.** The winding kernel has dimension 10 on Λ_8 and the kernel-diagonal values are $\{1/2, 3/4\}$. Any modification of the spectrum of H_{wind} that changes these values would falsify the framework.
6. **Dark energy.** The dark energy density is $\rho_{DE} = ((d+1)/2)(1+c\delta)m_\nu^4$, matching observation to 0.010%. Any future measurement of ρ_{DE} that deviates from this formula at more than 0.1% would falsify the framework.
7. **Charge rigidity.** The charge assignment on the 14-mode multiplet satisfies exactly two structurally consistent configurations, both with total charge $-1e$. A measurement of the vacuum's total electromagnetic charge that differs from $-1e$ (per one Standard Model generation) would falsify the rigidity theorem.
8. **Identification.** The Midpoint Sorting Theorem (HCSM-23 Rev. 7 [23]) assigns the 14 modes uniquely. A Standard Model particle with an $N/4$ label outside the set $\{60, 24, 21, 20, 14, 14, 12, 9.5, 9\}$ would falsify the framework.
9. **A_1 uniqueness.** The physical color algebra is the $\mathfrak{su}(3)$ subalgebra on the A_1 isotypic block, forced by the dimension-and-invariance theorem. A different embedding of $\mathfrak{su}(3)_c$ in the D_4 commutant would falsify Theorem 5.4 of HCSM-06 [6].
10. **Hypercharge values.** The hypercharge values are uniquely determined by the content structure and anomaly cancellation. A different set of hypercharge values satisfying the same content structure and anomaly conditions would falsify Theorem 8.1 of HCSM-07 [7].
11. **Mirror cancellation.** The Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$ and the residual ratio is exactly $7/32$. Any deviation from these exact structural identities would falsify the mirror-cancellation theorem (HCSM-10 [10]).
12. **TF grading and Klein orbit.** The TF operator grades the 14-mode multiplet by $|\Delta n|$, and the four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful Klein four-group action. Any modification of these exact structural identities would falsify HCSM-10 [10].
13. **5 + 9 split.** The 14-mode multiplet splits into exactly 5 Substrate Eigenmodes and 9 Mesh Resonances. Any modification of the derived base formula or shell values would falsify HCSM-11 [11].
14. **Layer independence.** The 14-mode multiplet carries four independent structural layers; layers 1, 2, 3 are HCSM-native and mutually compatible, while layer 4 requires external calibration. Any HCSM-native derivation of the processing layer shape parameters r_a would falsify the layer-independence theorem (HCSM-11 [11]).
15. **Entropic flatness.** The natural state on the 56-dimensional Hodge sector is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$, with entropy reduction $\Delta S = \log(7/5)$. Any deviation from these exact structural identities would falsify the Hodge-HCSM interface (HCSM-12 [12]).
16. **Loop and DNLS commutants.** The loop commutant is 3-dimensional abelian; the DNLS commutant is 10-dimensional, $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$. Any deviation from these dimensions would falsify HCSM-15 [15].
17. **Doubled-momentum pairing.** The B_1 -compressed kernel vanishes exactly: $K_{B_1} = 0$. Any nonzero value would falsify Theorem 5.30 of this paper.

18. **Chirality-breaking localization.** The commutator $[T, P_{rs}]$ is supported exactly on $\{\text{dark}, \tau\}$. Any additional support would falsify HCSM-13 [13].

10 Theorem Inventory

The framework's theorems are catalogued by their home paper. New theorems from the present revision cycle are marked with $\star$.

Theorem	Location
Laplacian uniqueness	HCSM-01 Thm 2.1
$d \cdot 2^d = 64$, $L = 8$	HCSM-01 Cor 4.1
Two derivations of $L = 8$	HCSM-01 §4, HCSM-14.b
$2(L - 1) = 14$	HCSM-01 Thm 4.4
Unified substrate selection	HCSM-00 §2.6
Single lattice primitive $\star$	HCSM-00 §2.7
Mirror symmetry $\star$	HCSM-00 Thm 2.16
$50 + 14$ decomposition $\star$	HCSM-00 Cor 2.19
Residual ratio $7/32$ $\star$	HCSM-00 Thm 2.22
TF grading by $ \Delta n $ $\star$	HCSM-00 Thm 2.28
Klein four-orbit $\star$	HCSM-00 Thm 2.32
Hodge-Dirac square	HCSM-02 Thm 3.2
Lorentzian signature	HCSM-02 Thm 3.9
Kernel dimension 4	HCSM-02 Thm 6.1
Spectral gap $2 \sin(\pi/L)$	HCSM-02 Thm 6.4
D_4 decomposition of V_{56}	HCSM-02 Thm 7.1
Self-paired eigenvalue	HCSM-04 Thm 3.2
Explicit mode list	HCSM-04 Thm 4.1
D_4 orbit decomposition	HCSM-04 Thm 5.1
Support classes $\{64, 48, 32\}$	HCSM-04 Prop 6.2
IPRs $\{2/128, 3/128, 4/128\}$	HCSM-04 Prop 7.2
Affine entropy-IPR relation	HCSM-04 Thm 8.2
Entropy classes $\{6.0, 5.5, 5.0\}$	HCSM-04 Cor 8.3
D_4 character χ_{14}	HCSM-05 Thm 4.1
Irreducible decomposition	HCSM-05 Thm 6.1
Joint (D, T) class partition $\star$	HCSM-05 Thm 7.7
Isotypic projector degeneracy (loop) $\star$	HCSM-15 Thm 4.1
No root generators (loop) $\star$	HCSM-15 Thm 4.3
A_1 uniqueness for color	HCSM-06 Thm 5.4
B_1/B_2 outer automorphism	HCSM-06 Thm 5.10
Sector-coarse 15-dim space	HCSM-06 Thm 6.3
A_1 hypercharge coefficients	HCSM-07 Thm 5.1
B_1 hypercharge coefficients	HCSM-07 Thm 6.1
Hypercharge quantization	HCSM-07 Thm 7.1
Hypercharge values from anomaly	HCSM-07 Thm 8.1
Content structure from HCSM data	HCSM-07 Thm 8.2
Orientation-flip $\mathbb{Z}_2$	HCSM-09 Thm 5.1
Sector-operator transformation	HCSM-09 Thm 6.1
Mirror cancellation	HCSM-10 Thm 3.4
Residual ratio $7/32$	HCSM-10 Thm 7.1
Convention independence	HCSM-10 Thm 8.1

Theorem	Location
Derived base formula*	HCSM-11 Thm 3.1
$5 + 9$ split*	HCSM-11 Thm 4.2
Shell formula*	HCSM-11 Thm 5.1
W/Z base formula*	HCSM-11 Thm 6.1
Mirror residual as Hartree eigenvalue*	HCSM-11 Thm 7.2
T_{half} commutes with Hartree*	HCSM-11 Thm 7.3
Coefficient 26*	HCSM-11 Thm 8.1
Layer independence*	HCSM-11 Thm 9.1
Entropic flatness	HCSM-12 Thm 3.1
Processor channel output	HCSM-12 Thm 4.2
Entropy reduction $\log(7/5)$	HCSM-12 Cor 5.1
Fermion/Boson-Class Correspondence	HCSM-13 Thm 3.1
Chirality-Breaking Localization*	HCSM-13 Thm 4.1
Observer pair (W, t) *	HCSM-13 §5.3
Kernel-Diagonal Theorem	HCSM-14 Thm 7.3
Kernel- T decomposition*	HCSM-14 Thm 8.1
Family-block eigenvalues*	HCSM-14 Thm 10.1
Four kernel constants*	HCSM-14 Thm 11.1
Rank formula*	HCSM-14 Thm 9.3
Loop commutant*	HCSM-15 Thm 3.3
Isotypic projector degeneracy*	HCSM-15 Thm 4.1
No root generators*	HCSM-15 Thm 4.3
DNLS commutant*	HCSM-15 Thm 5.4
Doubled-momentum pairing*	HCSM-15 Thm 6.2
Exact $\mathfrak{su}(2)_L$ preservation*	HCSM-15 Cor 6.3
$\mathfrak{su}(3)_c$ breaking*	HCSM-15 Thm 7.1
Explicit B_1 basis vectors*	HCSM-15 App A
Loop commutant universality	HCSM-16 Thm 10
Hidden-sector neutrality	HCSM-16 Thm 11
DNLS commutant universality	HCSM-16 Thm 15
Cartan rotation angles	HCSM-17 Thm 6.2
DNLS on torus	HCSM-18 Thm 4
Class-homogeneity of K	HCSM-18 Thm 14
IPR diagonal of K	HCSM-18 Thm 11
Doubled-momentum pairing	HCSM-18 Thm 17
$(d - 1)/2$ coefficient	HCSM-19 Thm 6.1
Hartree orbit trace identity	HCSM-19 Thm 7.1
Zero state uniqueness	HCSM-22 Thm 4
Zero state vector	HCSM-22 Thm 7
rs -oddness of zero state	HCSM-22 Thm 8
Exact ratios of zero state	HCSM-22 Thm 12
Zero-state selection principle	HCSM-22a Thm 3.1
Landauer route to selection	HCSM-22a Thm 4.1
Entropy-well route to selection	HCSM-22a Thm 5.1
Route equivalence	HCSM-22a Thm 6.1
Midpoint Sorting Theorem	HCSM-23 Thm 5.2
Within-pair ordering rule	HCSM-23 Thm 4.3
ν/e resolution	HCSM-23 Cor 4.4
Charge rigidity class	HCSM-24 Thm 6.3
Charge assignment solutions	HCSM-24 Thm 6.4

Theorem	Location
Total charge rigidity	HCSM-24 Thm 6.5
Mass map	HCSM-33 Thm 3.1
Geometric-mean identity	HCSM-33 Thm 5.1

All seven rank blockers of the framework’s derivation program are closed. As of the present revision, the physical identification of the A_1 and B_1 blocks is derived (color) or a labeling convention (weak isospin); the hypercharge values and content structure are derived; the mirror cancellation, $5 + 9$ split, Hodge-HCSM interface, orientation-flip $\mathbb{Z}_2$, processor commutant structure, and doubled-momentum pairing are integrated. The framework has zero empirical anchors in the particle, gauge, and hypercharge sectors.

11 Reproducibility

All numerical claims in the HCSM paper stack are verified at machine precision in IEEE-754 double precision. The verification scripts use only standard libraries (NumPy, SciPy, SymPy, mpmath) and construct the objects they need from HCSM-native definitions. Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the stack.

The complete HCSM paper stack, verification scripts, and source are available at

<https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM>

12 Conclusion

This paper has established the foundation of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by the unified substrate selection theorem (Section 2.6). The substrate carries the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation. The processor T and its five-layer composition $M_{5\text{layer}}$ are the machine’s operation. The output structures (charge, mass, flavor, gravity, cosmology) are the residue after mirror cancellation.

The framework’s most consequential structural results are now eleven.

First, the Kernel-Diagonal Theorem unifies the framework’s three leading coefficients under a single projector: the EFT coefficient $c = 5/9$, the mass-map coefficient $(d - 1)/2 = 3/2$, and the kernel dimension $10 = (d - 1)^2 + 1$ are three traces of the diagonal of the projector onto $\ker(H_{\text{wind}})$.

Second, the Λ -promotion mechanism is closed: the residual vacuum energy is promoted by pure projection onto the timelike direction of $\ker D$, and the hidden $\mathfrak{u}(1)$ VEVs vanish.

Third, the framework’s coverage of the Standard Model gauge content (absence of a light dark photon, spin-chirality $\mathbb{Z}_2$ grading, strong-CP solution, Proca sector, topological protection of ϕ_{gen}) is complete.

Fourth, the charge sector is a rigidity-class result: five structural constraints reduce the charge assignment from 5^{14} configurations to exactly two, both with total charge $-1e$, and the residual binary choice is resolved by the Midpoint Sorting Theorem.

Fifth, the identification thesis is closed: the 14 modes are uniquely sorted into the 14 Standard Model particles by native HCSM data, with zero empirical anchors remaining in the particle sector.

Sixth, the mirror-cancellation theorem (HCSM-10 [10]) is integrated into the foundations: the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ pairs 50 of the 64 scalar modes into cancelling partners, only the 14 self-paired modes at $\lambda = 4$ survive, and the residual ratio is exactly $7/32$.

Seventh, the $5 + 9$ split of the 14-mode multiplet (HCSM-11 [11]) and the Hodge-HCSM interface (HCSM-12 [12]) are integrated: 5 Substrate Eigenmodes and 9 Mesh Resonances; the natural state on the 56-dimensional Hodge sector is the maximally mixed state; the processor channel is a completely positive trace-preserving map; and the entropy reduction is exactly $\log(7/5)$.

Eighth, the processor commutant structure (HCSM-15 [15]) is integrated: the loop commutant is a 3-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$; the DNLS commutant is 10-dimensional, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$.

Ninth, the doubled-momentum pairing mechanism (HCSM-15 [15]) is integrated: the B_1 -compressed Hartree kernel vanishes identically, $K_{B_1} = 0$, by the explicit B_1 basis vectors constructed in Appendix A.

Tenth, the chirality-breaking localization (HCSM-13 [13]) is integrated: the processor commutator $[T, P_{rs}]$ is supported exactly on the two-element set $\{\text{dark}, \tau\}$.

Eleventh, the winding generator kernel structure (HCSM-14 [14]) is refined: the kernel- T decomposition is $(6, 1, 3)$, the family-block eigenvalues are $\pm L/2$, and the four parity-block indicators span a 4-dimensional subspace of the kernel.

The present revision closes the physical identification of the gauge and hypercharge sectors. The A_1 isotypic block is uniquely identified with physical color by a dimension-and-invariance theorem (HCSM-06 [6], Theorem 5.4). The B_1 block is identified with physical weak isospin up to the outer-automorphism labeling convention (HCSM-06 [6], Theorem 5.10). The Standard Model content structure and hypercharge values are derived from HCSM-native data and anomaly cancellation (HCSM-07 [7], Theorems 8.1 and 8.3). The framework now has zero empirical anchors in the particle, gauge, and hypercharge sectors.

The framework has a single dimensionful anchor (the Planck mass M_P) and derives every other quantity from the substrate at $d = 4$ and $L = 8$.

The framework's claim is that the Standard Model is what the substrate looks like when you observe it.

Physics is what the machine cannot cancel. The Hodge complex is the machine. The $-1e$ is what the machine must not cancel.

A Explicit B_1 Basis Vectors and the Doubled-Momentum Pairing Theorem

This appendix constructs the explicit orthonormal basis vectors of the B_1 isotypic sector of the 14-mode multiplet and verifies the doubled-momentum pairing property used in Theorem 5.30.

A.1 The B_1 isotypic projector

The B_1 isotypic projector on the 14-mode multiplet is

$$P_{B_1} = \frac{1}{8} \left[R(e) + R(r^2) - R(r) - R(r^3) + R(s) + R(r^2s) - R(rs) - R(r^3s) \right], \quad (117)$$

where the D_4 action on mode coordinates is given by (4). The B_1 character is $\chi_{B_1} = (1, 1, -1, 1, -1)$ on the conjugacy classes (e, r^2, r, s, rs) .

A.2 Explicit construction of the basis vectors

Proposition A.1 (First B_1 basis vector). *Applying P_{B_1} to the neutrino mode $\nu = (0, 4)$ gives, after normalization,*

$$v_1^{(B_1)} = \frac{|\nu\rangle - |e\rangle}{\sqrt{2}} = \frac{|(0, 4)\rangle - |(4, 0)\rangle}{\sqrt{2}}. \quad (118)$$

Proof. Direct computation:

$$\begin{aligned} P_{B_1}|\nu\rangle &= \frac{1}{8} \left[|\nu\rangle + |\nu\rangle - |e\rangle - |e\rangle + |\nu\rangle + |\nu\rangle - |e\rangle - |e\rangle \right] \\ &= \frac{1}{8} \left[4|\nu\rangle - 4|e\rangle \right] = \frac{|\nu\rangle - |e\rangle}{2}. \end{aligned} \quad (119)$$

Normalizing gives (118). $\square$

Proposition A.2 (Second B_1 basis vector). *Applying P_{B_1} to the up-quark mode $u = (1, 3)$ gives, after normalization,*

$$v_2^{(B_1)} = \frac{\sqrt{2}}{4} \left(|u\rangle + |\tau\rangle - |b\rangle - |\mu\rangle + |d\rangle + |c\rangle - |s\rangle - |\text{dark}\rangle \right). \quad (120)$$

In coordinate form:

$$\begin{aligned} v_2^{(B_1)} &= \frac{\sqrt{2}}{4} \left(|(1, 3)\rangle + |(7, 5)\rangle - |(5, 1)\rangle - |(3, 7)\rangle \right. \\ &\quad \left. + |(1, 5)\rangle + |(7, 3)\rangle - |(3, 1)\rangle - |(5, 7)\rangle \right). \end{aligned} \quad (121)$$

Proof. Direct computation of $P_{B_1}|u\rangle$ gives

$$P_{B_1}|u\rangle = \frac{1}{8} \left[|u\rangle + |\tau\rangle - |b\rangle - |\mu\rangle + |d\rangle + |c\rangle - |s\rangle - |\text{dark}\rangle \right]. \quad (122)$$

The eight coefficients are $\pm 1/8$ in the standard mode order. Normalizing: $\|P_{B_1}|u\rangle\|^2 = 8 \cdot (1/8)^2 = 1/8$, so the normalized vector is $\sqrt{8} \cdot P_{B_1}|u\rangle = \frac{\sqrt{2}}{4}(\dots)$. $\square$

Proposition A.3 (Orthonormality). *The two vectors $v_1^{(B_1)}$ and $v_2^{(B_1)}$ are orthonormal:*

$$v_1^{(B_1)} \cdot v_1^{(B_1)} = 1, \quad v_2^{(B_1)} \cdot v_2^{(B_1)} = 1, \quad v_1^{(B_1)} \cdot v_2^{(B_1)} = 0. \quad (123)$$

Proof. For v_1 : $\|v_1\|^2 = \frac{1}{2}(1+1) = 1$. For v_2 : $\|v_2\|^2 = \frac{2}{16} \cdot 8 = 1$. For the inner product: the supports of v_1 and v_2 are disjoint (v_1 is supported on $\{\nu, e\} \subset \mathcal{O}_0$, while v_2 is supported on $\{u, d, s, \mu, \text{dark}, c, \tau, b\} \subset \mathcal{O}_1$), so the inner product vanishes. $\square$

A.3 The matrix U_{B_1}

In the standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$:

$$U_{B_1} = \begin{pmatrix} 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 0 \\ 0 & \sqrt{2}/4 \\ 0 & \sqrt{2}/4 \\ 0 & -\sqrt{2}/4 \\ 0 & -\sqrt{2}/4 \\ 0 & -\sqrt{2}/4 \\ 0 & \sqrt{2}/4 \\ 0 & \sqrt{2}/4 \\ 0 & -\sqrt{2}/4 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad (124)$$

This matrix satisfies $U_{B_1}^T U_{B_1} = \mathbb{I}_2$.

A.4 Verification of the B_1 transformation properties

Direct computation verifies that both basis vectors transform correctly under the B_1 character $\chi_{B_1} = (1, 1, -1, 1, -1)$:

g	$R(g)v_1$	$R(g)v_2$
e	$+v_1$	$+v_2$
r^2	$+v_1$	$+v_2$
r	$-v_1$	$-v_2$
r^3	$-v_1$	$-v_2$
s	$+v_1$	$+v_2$
r^2s	$+v_1$	$+v_2$
rs	$-v_1$	$-v_2$
r^3s	$-v_1$	$-v_2$

Both vectors transform as B_1 . This confirms that U_{B_1} is a valid isotypic basis.

A.5 The doubled-momentum pairing structure

Proposition A.4 (Pairing structure of v_1). *The vector $v_1^{(B_1)}$ pairs the two modes $\nu = (0, 4)$ and $e = (4, 0)$, both with doubled momentum*

$$(2 \cdot 0, 2 \cdot 4) \equiv (2 \cdot 4, 2 \cdot 0) \equiv (0, 0) \pmod{8}, \quad (125)$$

and with opposite signs $+1/\sqrt{2}$ and $-1/\sqrt{2}$.

Proposition A.5 (Pairing structure of v_2). *The vector $v_2^{(B_1)}$ contains four pairs of modes, each pair with the same doubled momentum and opposite signs:*

Pair	Mode 1	Double mom	Sign	Mode 2	Double mom	Sign
1	$u = (1, 3)$	$(2, 6)$	+	dark = $(5, 7)$	$(2, 6)$	−
2	$\tau = (7, 5)$	$(6, 2)$	+	$s = (3, 1)$	$(6, 2)$	−
3	$d = (1, 5)$	$(2, 2)$	+	$b = (5, 1)$	$(2, 2)$	−
4	$c = (7, 3)$	$(6, 6)$	+	$\mu = (3, 7)$	$(6, 6)$	−

Proof. Direct computation from (121). For pair 1: $u = (1, 3)$ has doubled momentum $(2, 6)$; dark = $(5, 7)$ has doubled momentum $(10, 14) \equiv (2, 6) \pmod{8}$. The coefficients in v_2 are $+\sqrt{2}/4$ and $-\sqrt{2}/4$ respectively. The other three pairs are verified analogously. $\square$

A.6 Proof that $K_{B_1} = 0$

Theorem A.6 (B_1 -compressed kernel vanishes). *The B_1 -compressed Hartree kernel vanishes identically:*

$$K_{B_1} = U_{B_1}^T K U_{B_1} = 0 \in \mathbb{R}^{2 \times 2}. \quad (126)$$

Proof. The compression K_{B_1} has entries

$$(K_{B_1})_{ij} = (v_i^{(B_1)})^T K v_j^{(B_1)} = \sum_x \rho_i(x) \rho_j(x), \quad (127)$$

where the density profiles are

$$\rho_i(x) = \sum_a v_i^{(a)} |\phi_a(x)|^2, \quad i = 1, 2. \quad (128)$$

For $i = 1$: using Proposition A.4,

$$\rho_1(x) = \frac{1}{\sqrt{2}} (|\phi_\nu(x)|^2 - |\phi_e(x)|^2). \quad (129)$$

Since $\nu = (0, 4)$ and $e = (4, 0)$ are both in $\mathcal{O}_0$, we have $|\phi_\nu(x)|^2 = |\phi_e(x)|^2 = 1/64$ for all $x \in \Lambda$. Therefore $\rho_1(x) = 0$ for all $x \in \Lambda$.

For $i = 2$: using Proposition A.5,

$$\begin{aligned} \rho_2(x) = \frac{\sqrt{2}}{4} & \left[|\phi_u(x)|^2 - |\phi_{\text{dark}}(x)|^2 + |\phi_\tau(x)|^2 - |\phi_s(x)|^2 \right. \\ & \left. + |\phi_d(x)|^2 - |\phi_b(x)|^2 + |\phi_c(x)|^2 - |\phi_\mu(x)|^2 \right]. \end{aligned} \quad (130)$$

For each of the four pairs, the two modes have equal density: $|\phi_u(x)|^2 = |\phi_{\text{dark}}(x)|^2$ (by the doubled-momentum pairing), $|\phi_\tau(x)|^2 = |\phi_s(x)|^2$, $|\phi_d(x)|^2 = |\phi_b(x)|^2$, $|\phi_c(x)|^2 = |\phi_\mu(x)|^2$. Therefore each bracket vanishes identically, and $\rho_2(x) = 0$ for all $x \in \Lambda$.

Since $\rho_1 = \rho_2 = 0$ identically, every entry of K_{B_1} vanishes:

$$K_{B_1} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad (131)$$

□

A.7 Corollary: exact $\mathfrak{su}(2)_L$ preservation

Corollary A.7 ($\mathfrak{su}(2)_L$ preservation on the 14-mode multiplet). *The three generators of $\mathfrak{su}(2)_L$ commute with the Hartree kernel and the DNLS Hamiltonian:*

$$[T_L^i, K] = 0, \quad [T_L^i, H_{\text{DNLS}}] = 0, \quad i = 1, 2, 3. \quad (132)$$

Proof. Since $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ has support only on the B_1 isotypic sector:

$$[T_L^i, K] = U_{B_1} [\sigma^i, K_{B_1}] U_{B_1}^T + (\text{terms on non-}B_1 \text{ sectors that vanish by support}). \quad (133)$$

By Theorem A.6, $K_{B_1} = 0$, so $[\sigma^i, K_{B_1}] = 0$ identically. Hence $[T_L^i, K] = 0$ for all i . Since $[\sigma^i, \mathbb{I}] = 0$ trivially, $[T_L^i, H_{\text{DNLS}}] = 0$. □

B Web Crawler Metadata

B.1 Dublin Core

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    theorem, the 5+9 split, the Hodge-HCSM interface, the processor
    commutant structure, the doubled-momentum pairing mechanism, and the
    chirality-breaking localization into the foundations.}
}
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